Digital Image Processing and Computer Vision

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What is Digital Image Processing

- is any form of **Information Processing** for which the input is an image.
- such as photographs or frames of video.
- output is not necessarily an image, but can be for instance a set of features of the image.
- Most image-processing techniques involve treating the image as a **two-dimensional** signal and applying standard Signal-Processing techniques to it.

Digital Image Processing (Con't)

- **Input** – Image, Photograph, frame of video
- **Input Device** – Digital Camera, Video, Scanner, Image Construction Programs.
- **Process** – in This Class.
- **Output** – Image, Image Feature vector.

Digital Image Processing (Con't)

- Processing of a multidimensional pictures by a digital computer.
- การประมวลผลสัญญาณรูปภาพโดยใช้ดิจิตอลคอมพิวเตอร์.

Why we need Digital Image Processing?

- เพื่อการบันทึก และการจัดเก็บภาพ
- เพื่อปรับปรุงคุณภาพของภาพให้ดียิ่งขึ้น โดยการใช้กระบวนการทางคณิตศาสตร์ และสถิติเข้ามาช่วย
- เพื่อช่วยในการวิเคราะห์อะไรบางอย่างที่ซ้อนอยู่ในภาพรูปภาพ
- เพื่อสังเคราะห์ หรือสร้างภาพขึ้นมาเอง
- เพื่อสร้างระบบการมองเห็นให้กับคอมพิวเตอร์ หรือ ที่รู้จักกันในนามของ Computer Vision

Computer Vision

- Computer vision is the science and technology of machines that see.
- As a scientific discipline, computer vision is concerned with the theory and technology for building artificial systems that obtain information from images.
- The image data can take many forms, such as a video sequence, views from multiple cameras, or multi-dimensional data from a medical scanner.

Computer Vision (Con't)

- Controlling Processes.
  - Industrial Robot.
  - Autonomous Vehicle.
- Detecting Events.
  - Visual Surveillance.
- Organizing Information.
  - for indexing databases of images and image sequences.

Computer Vision (Con't)

- Modeling objects or environments
  - industrial inspection.
  - medical image analysis.
  - topographical modeling.
- Interaction
  - as the input to a device for computer-human interaction.
  - Tangible User Interface.
  - Surface Computing.

Image Terminology

- **Pixel** – Picture Cell, Picture Element.
- **Intensity** – Color Value in each pixel of Image.
- **Resolution** – Image Resolution, Display Screen Resolution.
- **RGB** – Red Green Blue.
- **Background**.
- **Foreground**.

Resolution

- How quality of image
- With the same size of picture
  - If high resolution, high memory is required to store data
  - If low resolution, less memory is required to store data
- Its unit is call “point per inch”

Resolution (Con't)

72

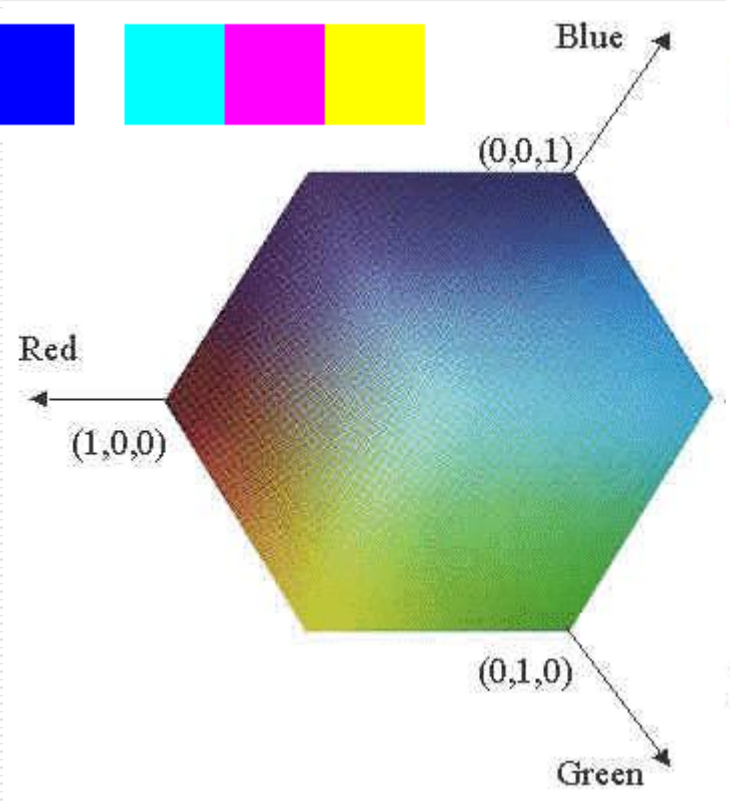
36

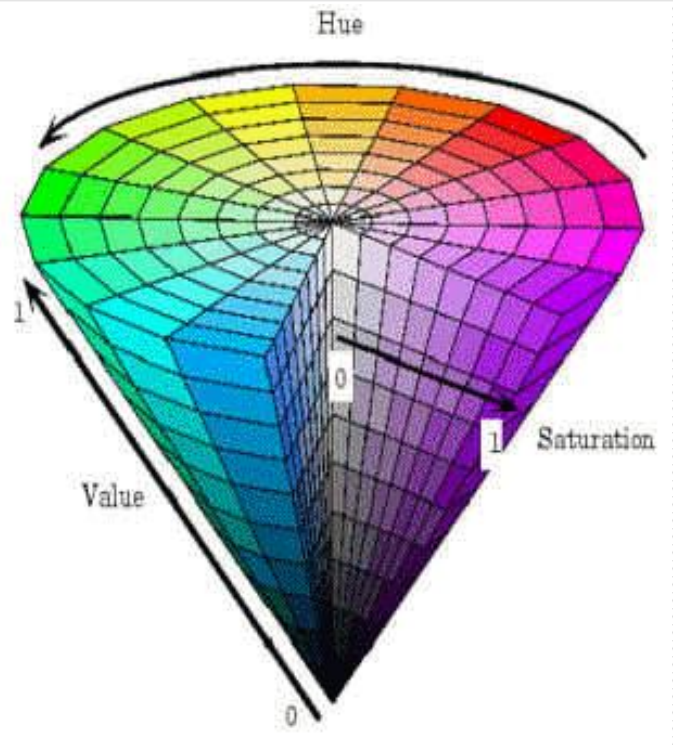
18

9

Color Models

- **RGB – CMYK - HSV**

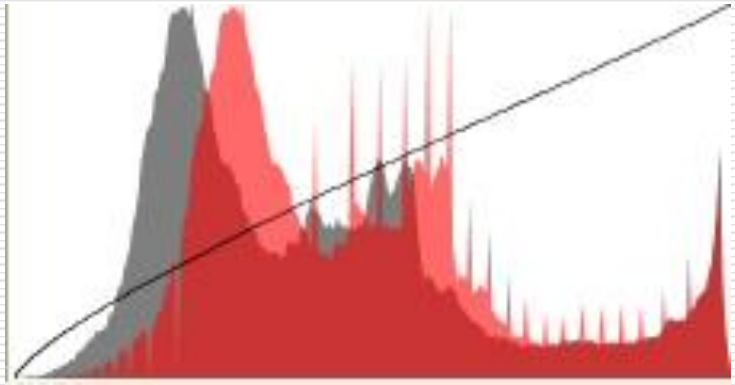




Color Models (Con’t)

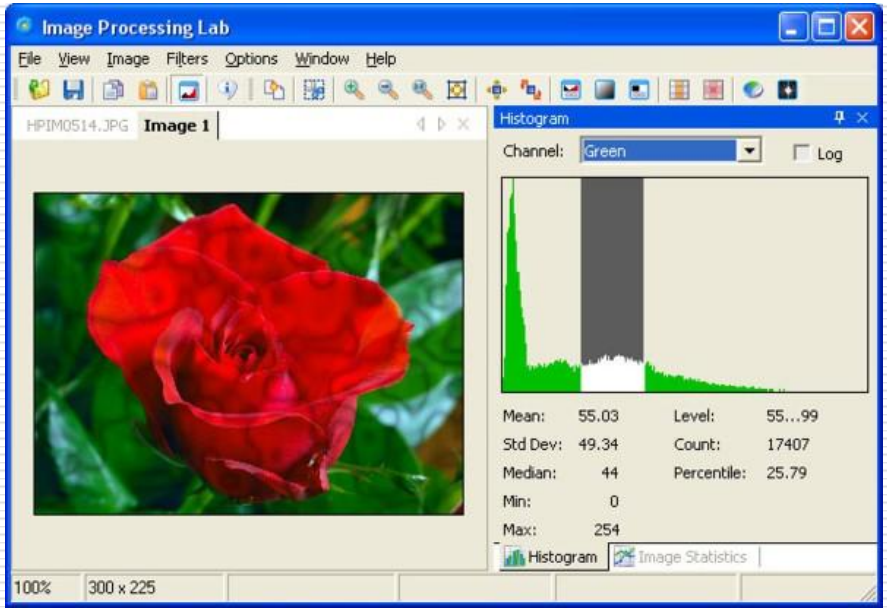
Histogram Processing

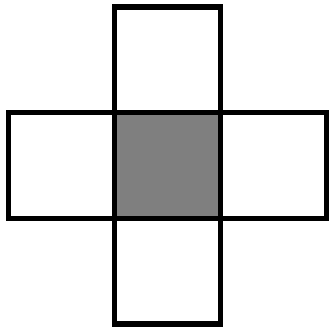
- Histogram.

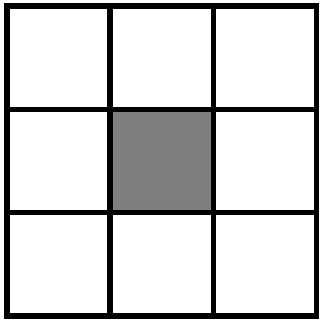


Histogram & Pixel Relation

- Color Histogram.
- 8nb, 4nb, 4nb orthogonal.







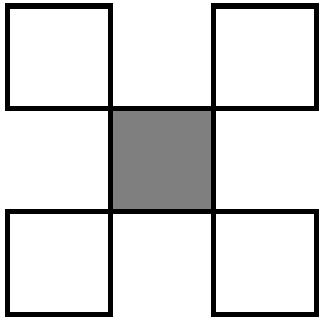


Image Enhancement

- Image Editing was known as photo retouching.

Image Enhancement (Con't)

- Cropping an Image.

Image Enhancement (Con't)

- Removal of unwanted elements.

Image Enhancement (Con't)

- Selective colour change.

Image Enhancement (Con't)

- Image orientation, Merging of images.

Image Enhancement (Con't)

- Special effects, Change colour depth.

Image Enhancement (Con't)

- Image Restoration.

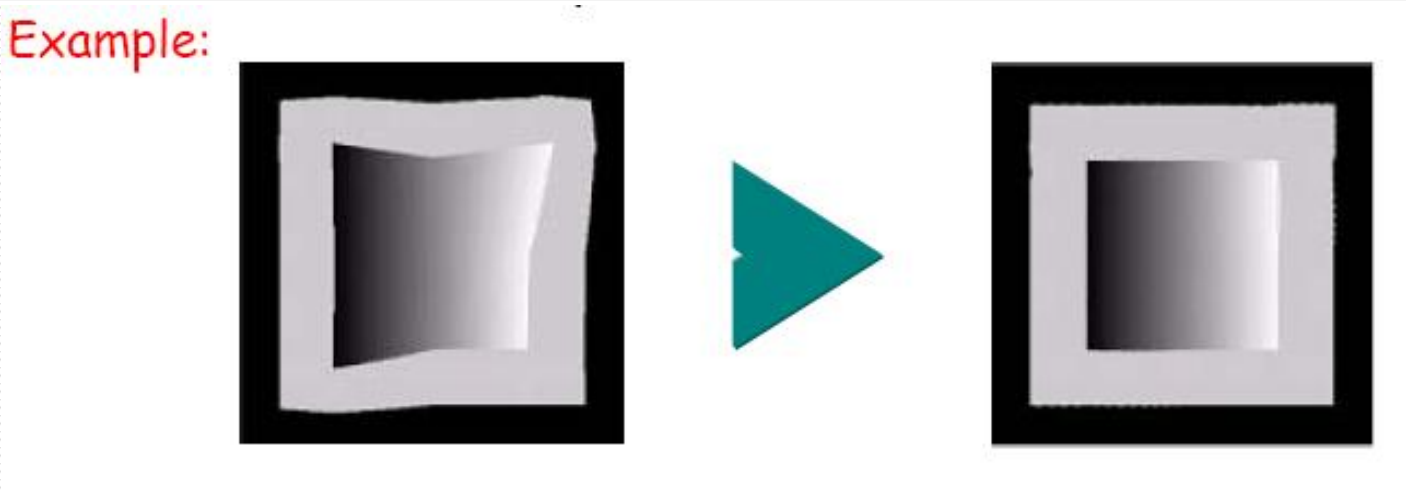


Image Enhancement (Con't)

- Image Restoration.

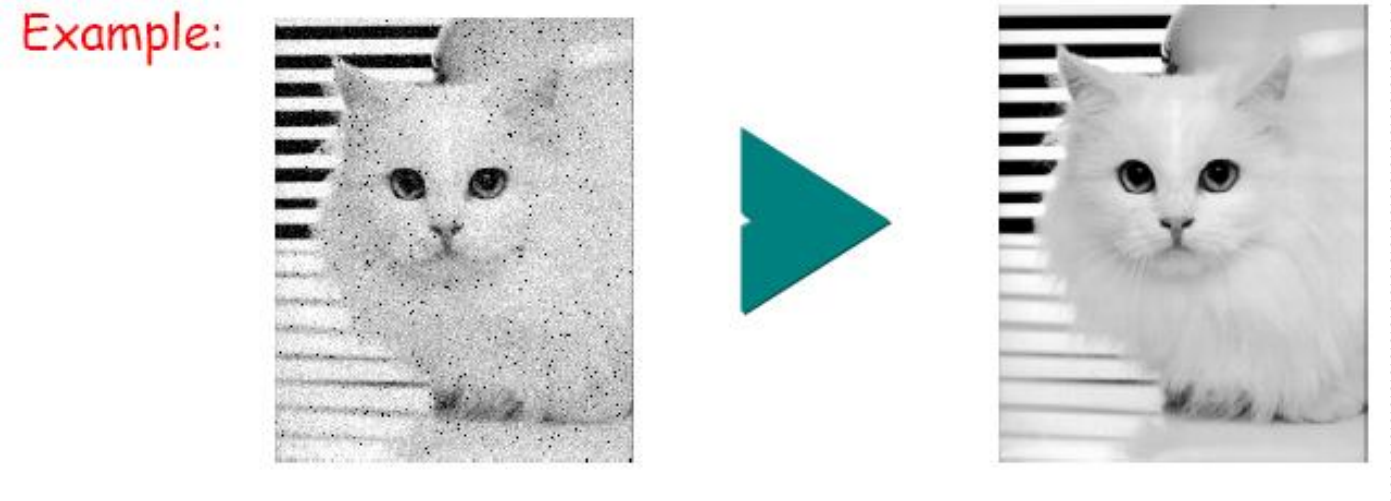


Image Enhancement (Con't)

- Image Restoration.

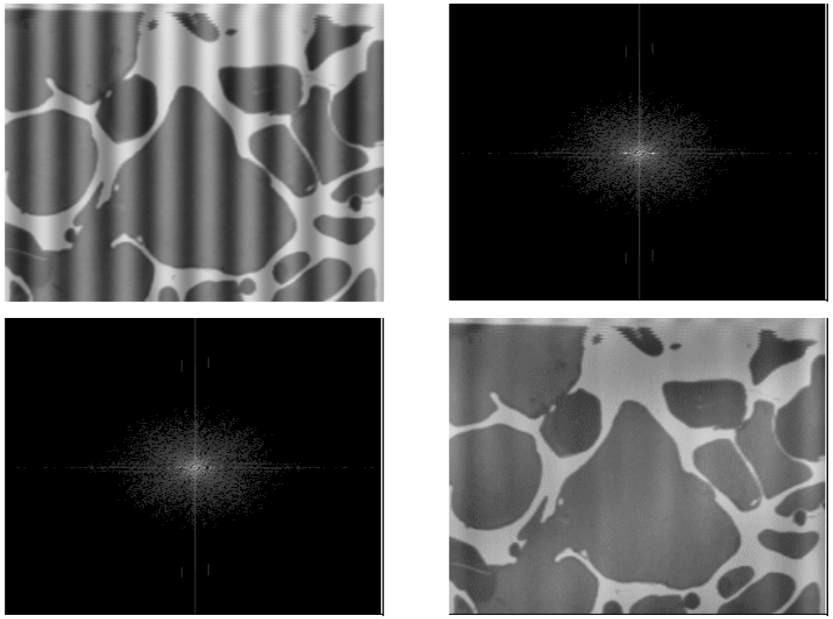


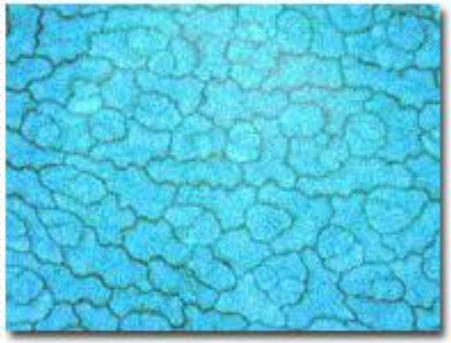
Figure 3. Application of the 2-dimensional Fast Fourier Transform

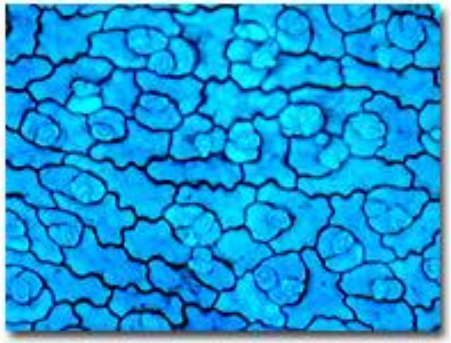
Figure 1. Application of the median filter

Figure 2. Adjusting the image histogram to improve image contrast

Image Equalization

- Contrast change and brightening.





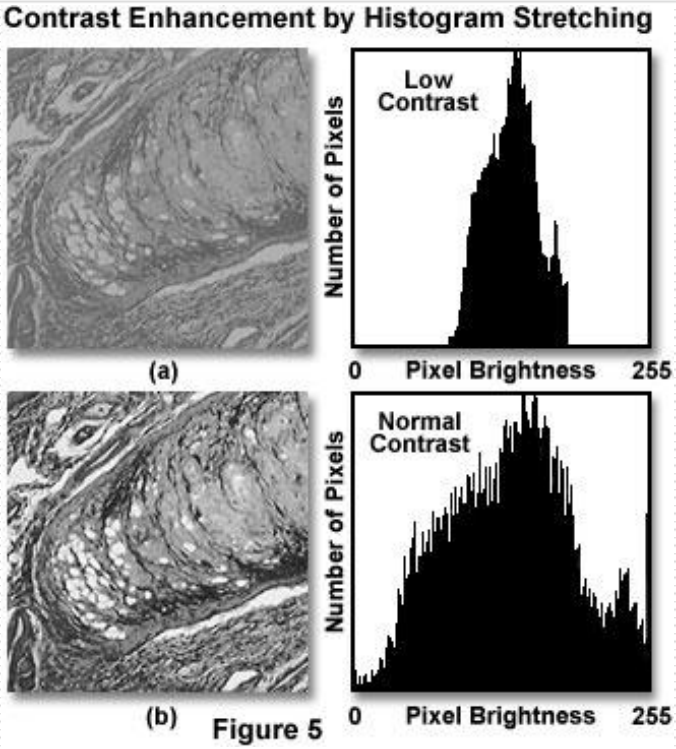


Image Equalization (Con't)

Image direct from camera

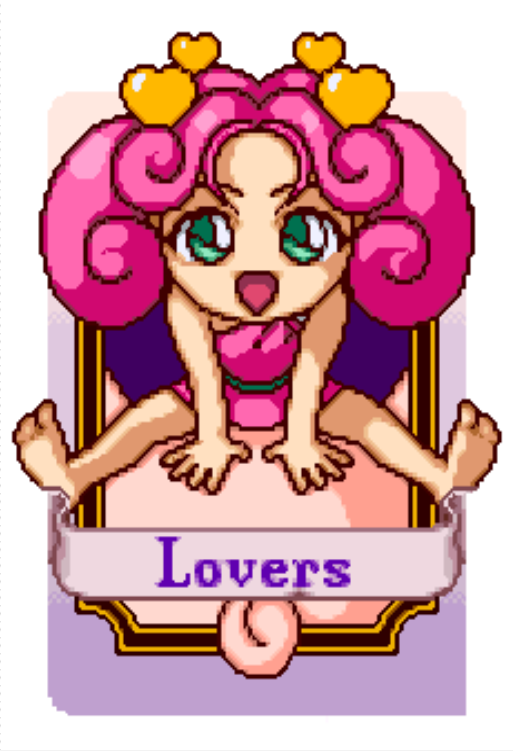
Processed image

Image Equalization (Con't)

Resized Image/Zoom Image

Linear Transform Function

- Inverted Image.

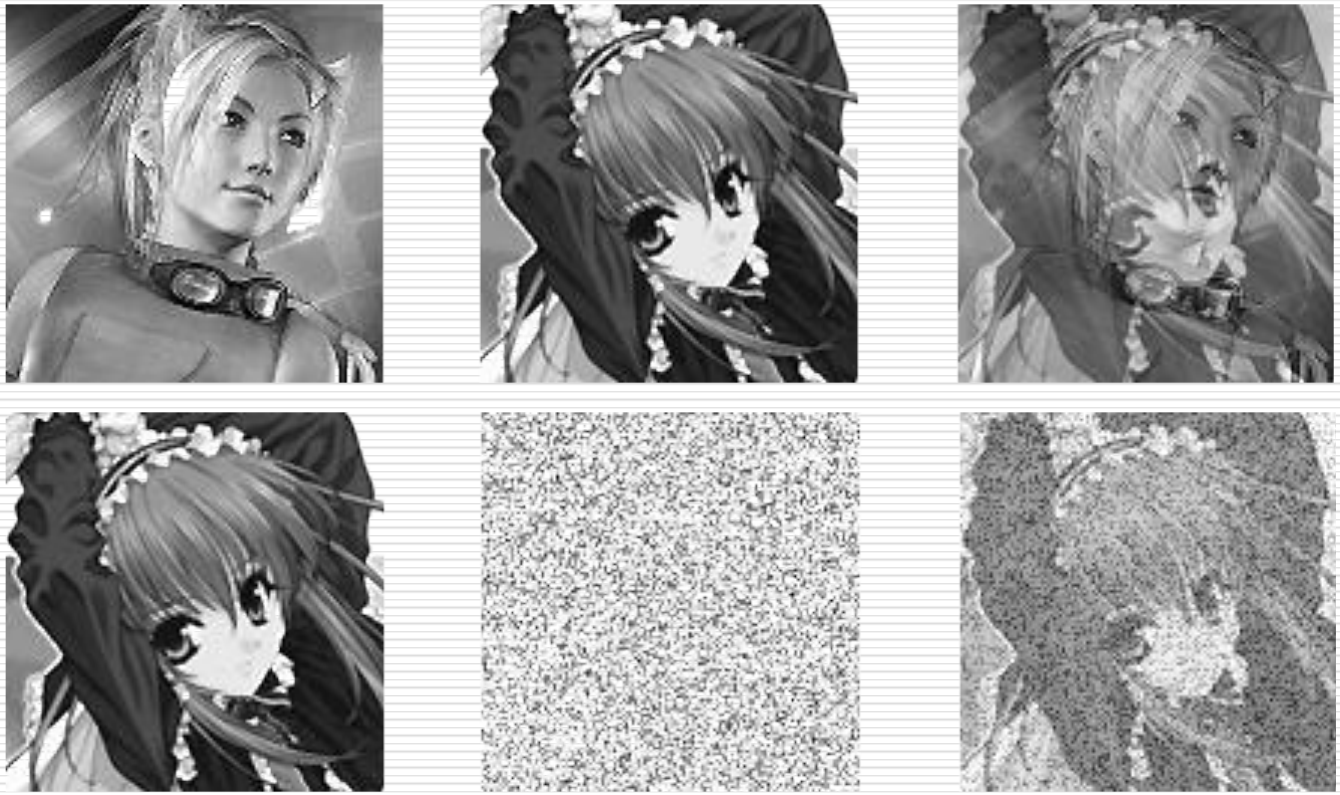


Linear Transform Function (Con't)

- Image Rotation.

Mathematical Operators

- Image Add Image (Pixel + Pixel).

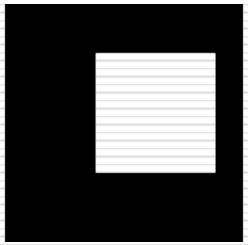


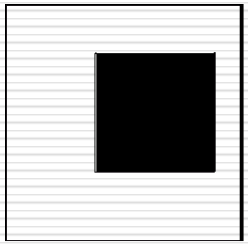
Mathematical Operators (Con't)

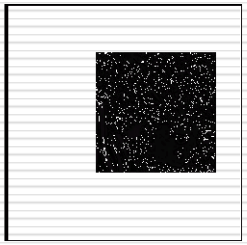
- Subtract, Multiply and Divide.

Logical Operators (Con't)

- AND, OR, NOT, XOR.







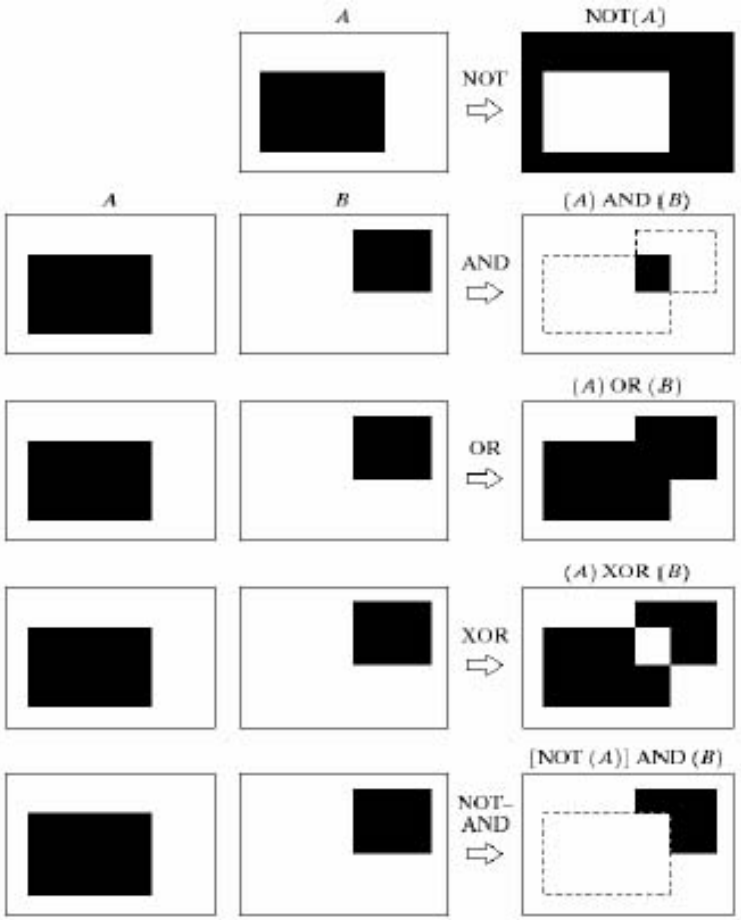
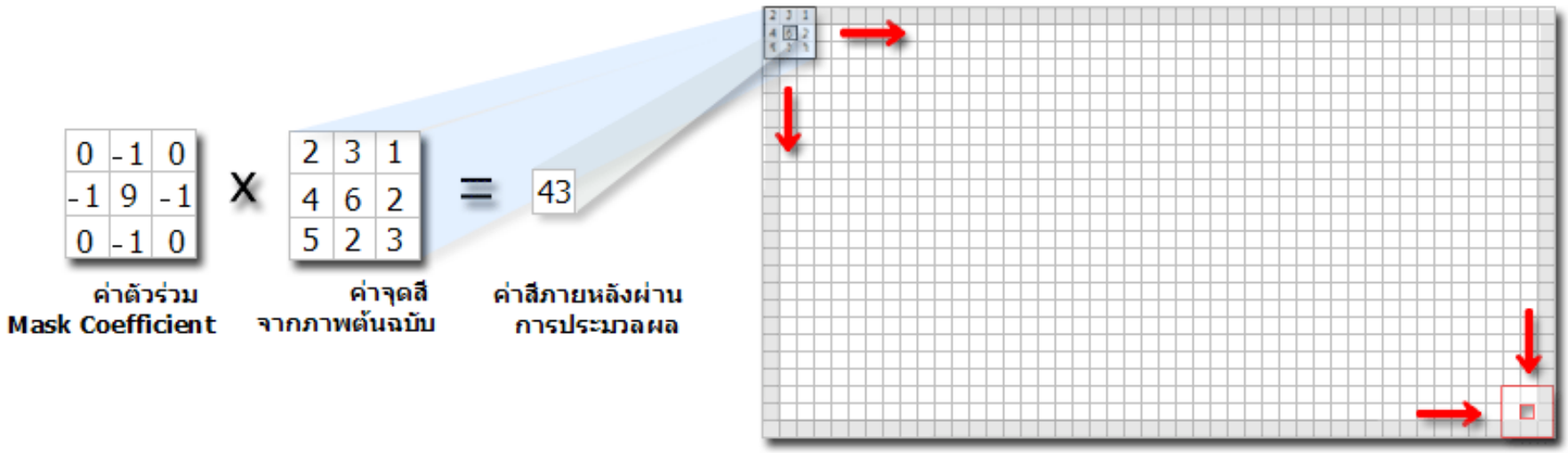


Image Convolution



Smoothing and Sharpening

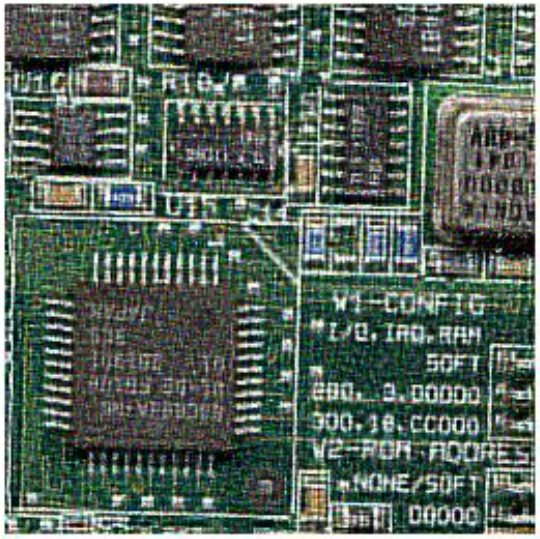
Smoothing and Sharpening

- การแก้ไขภาพที่เบลอ + มีสัญญาณรบกวน

Blurred & Noisy

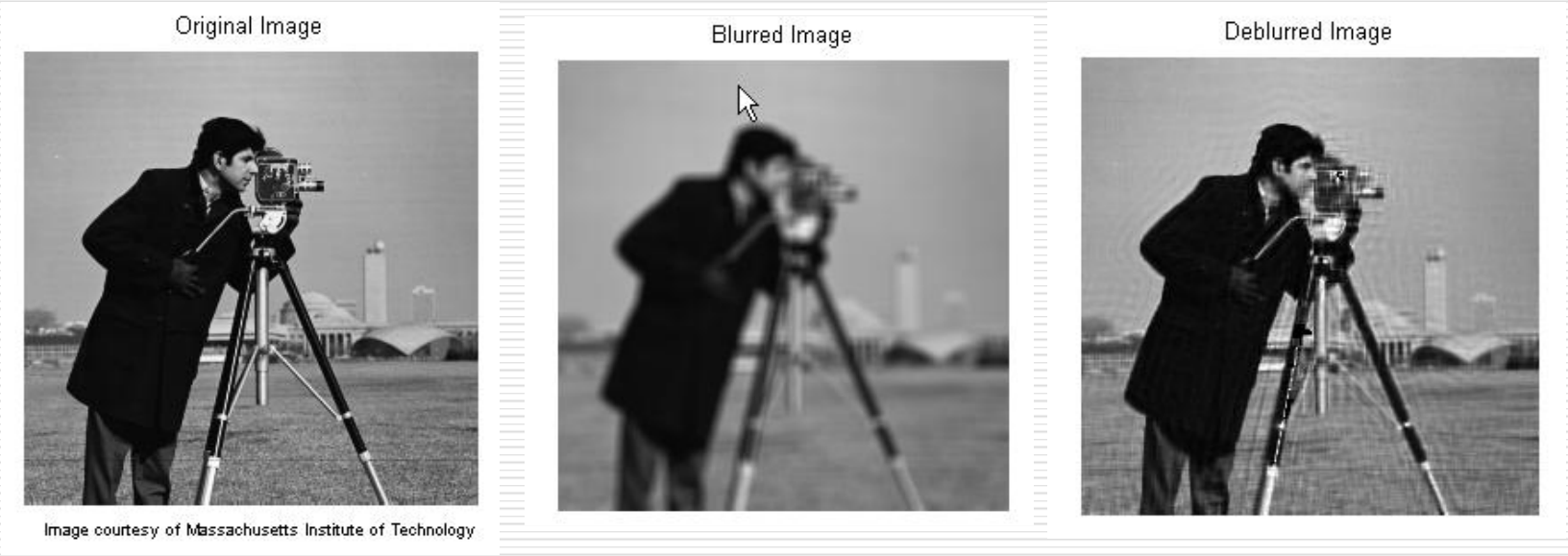
Restored Image, NUMIT = 5

Restored Image, NUMIT = 15

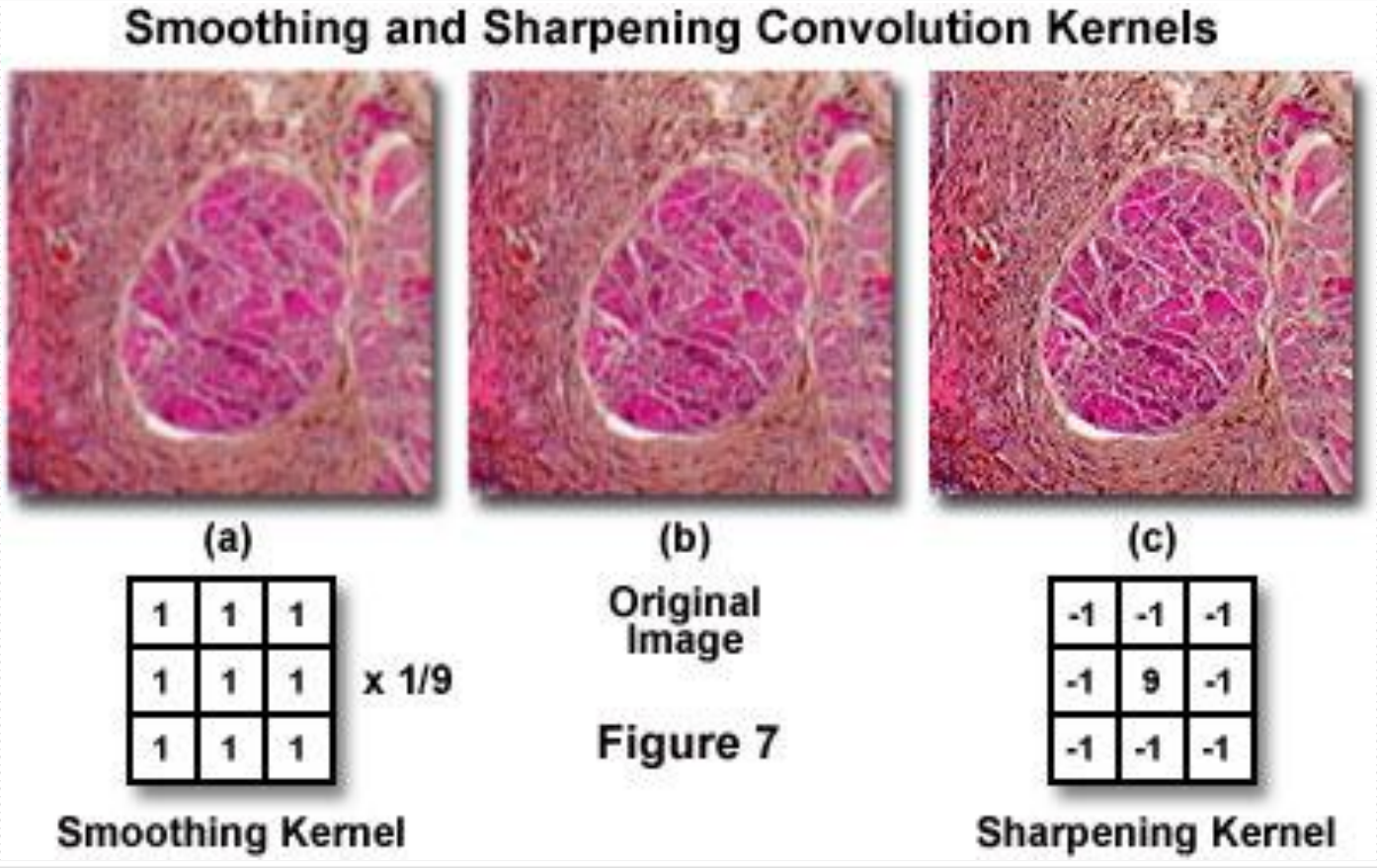


Smoothing and Sharpening

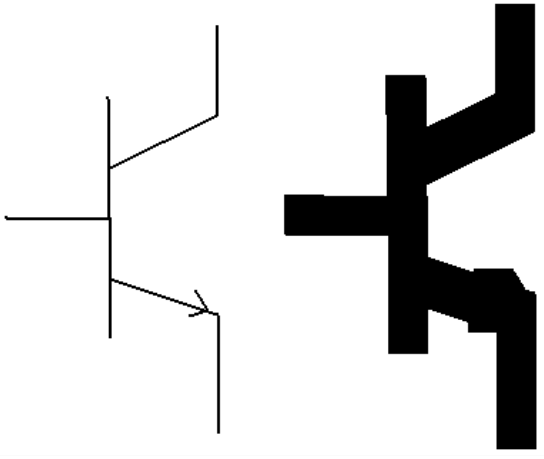
- การแก้ไขภาพที่เบลอ หรือการสร้างภาพเบลอ

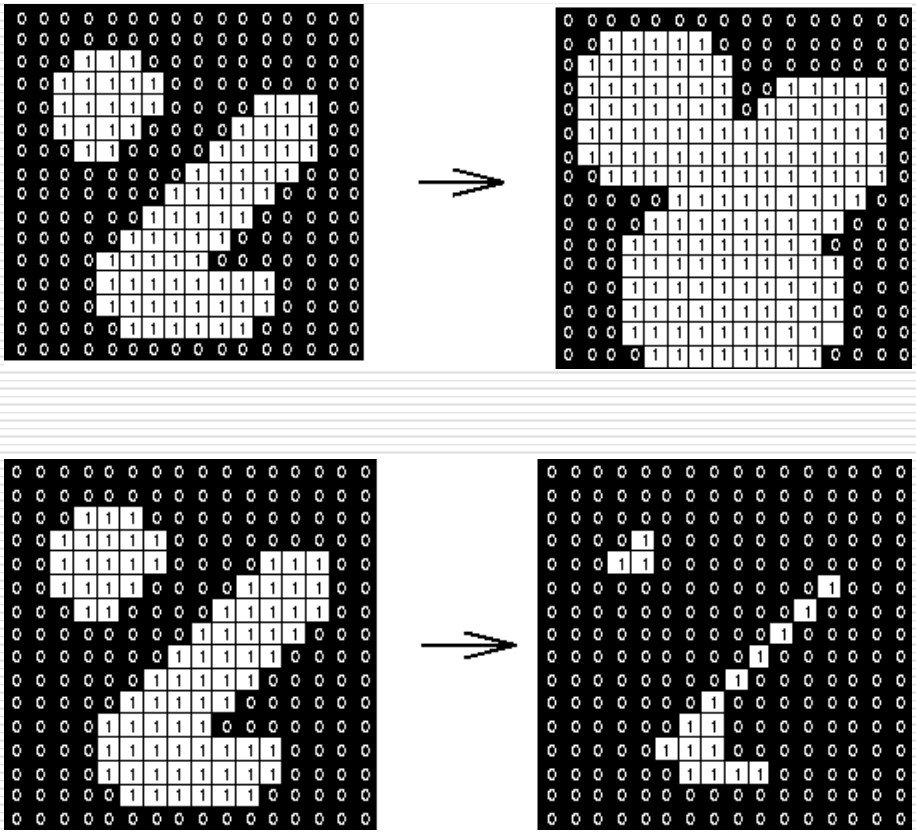


Smoothing and Sharpening



Dilation and Erosion

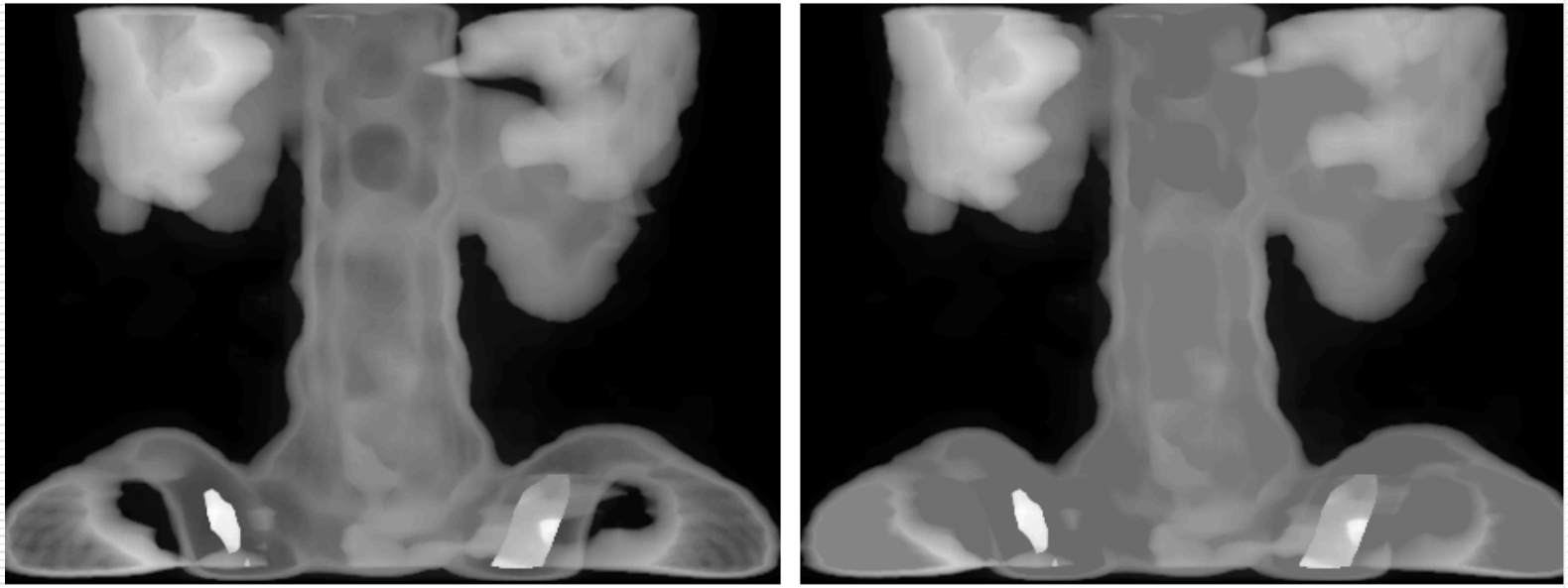




Dilation and Erosion (Con't)

- Dilation คือ การขยายภาพโดยมีสัดส่วนเท่ากันทั่วทั้งภาพ (Uniform)
- Erosion คือ การย่อภาพ ส่วนการทำ Skeleton เป็นการหาโครงสร้างหลักของวัตถุ

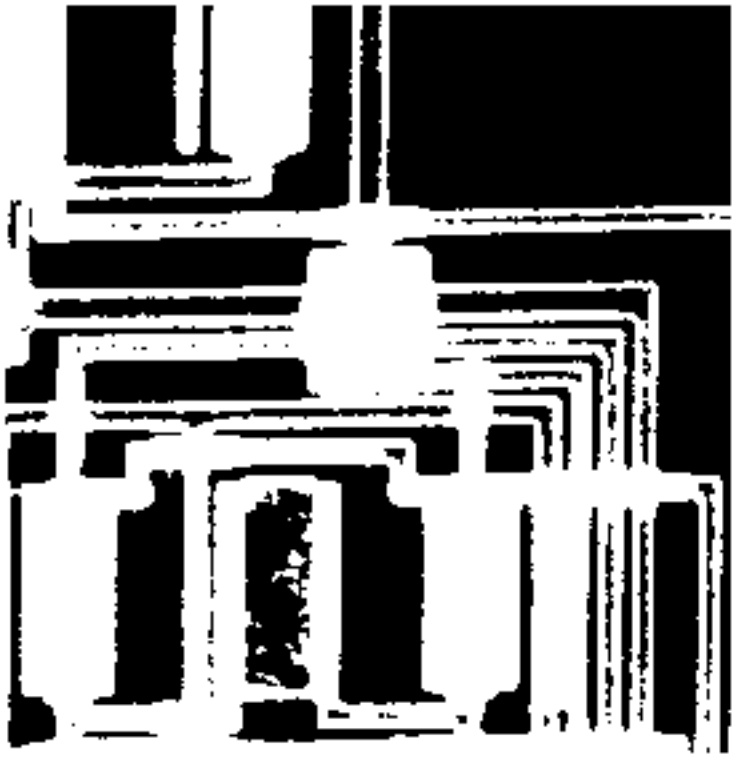
Dilation Application



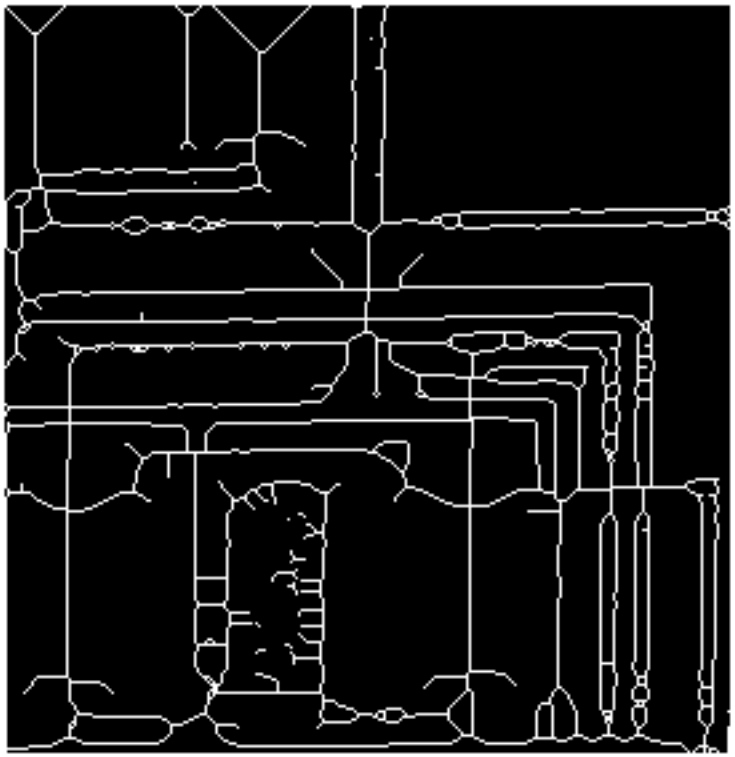
Original

After Filling Holes

Erosion Application



Original Image

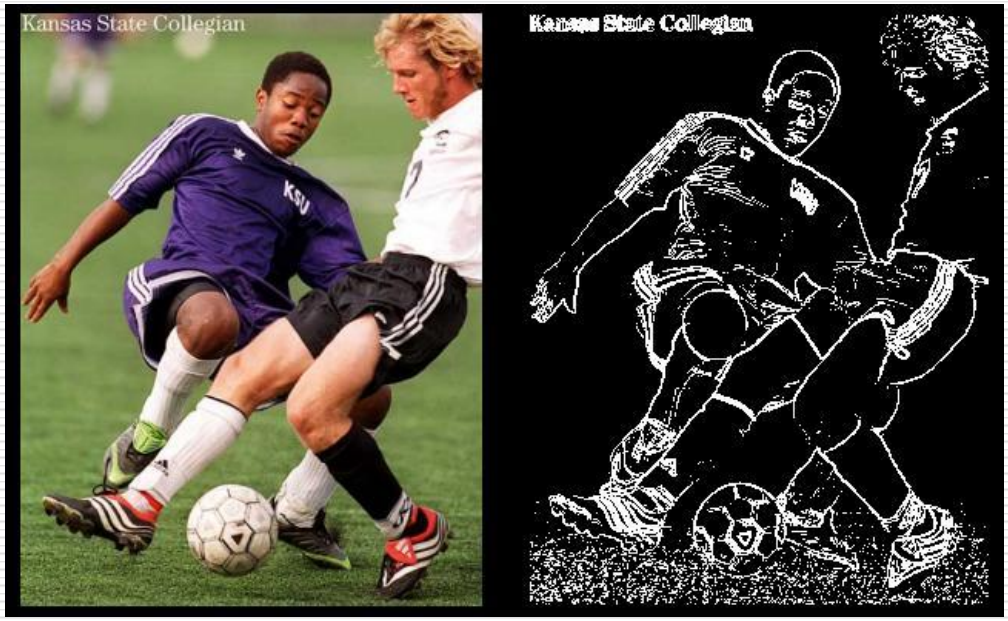


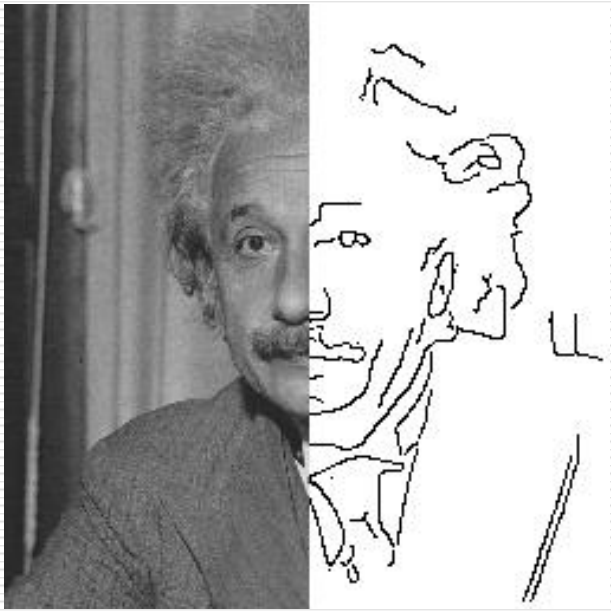
Skeletonization of Image

Image Segmentation

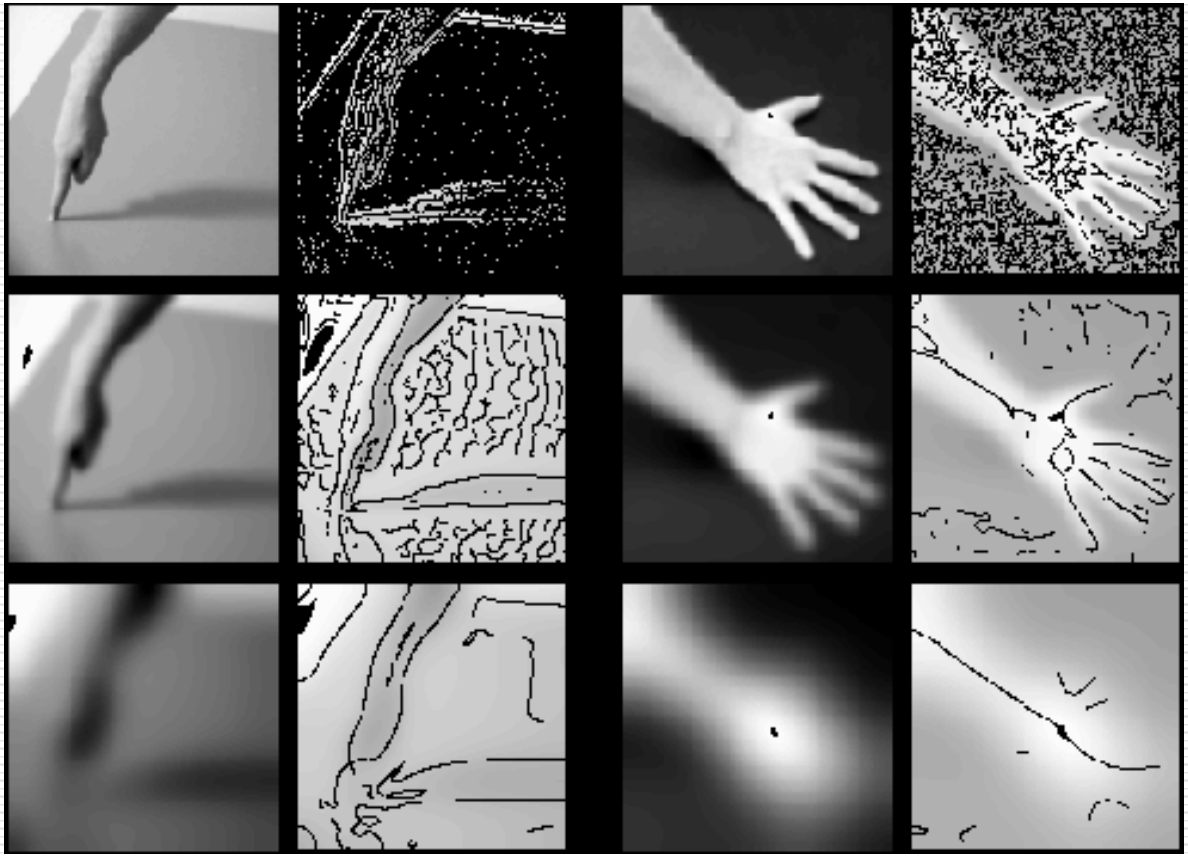
- เป็นกระบวนการตัดแยกภาพออกเป็นชิ้นส่วนย่อย โดยการตัดแยกวัตถุนั้นสามารถแบ่งออกได้เป็น 2 ชนิดดังนี้
  - การตัดแบ่งวัตถุตามเส้นขอบของวัตถุ (by identifying edges or lines) ได้แก่วิธีการของ Edge Detection, Line Detection
  - การตัดแบ่งวัตถุตามพื้นที่ของวัตถุ (by identifying regions or areas) ได้แก่วิธีการของ Hough Transform

Edge Detection





Edge Detection (Con't)



Edge Detection

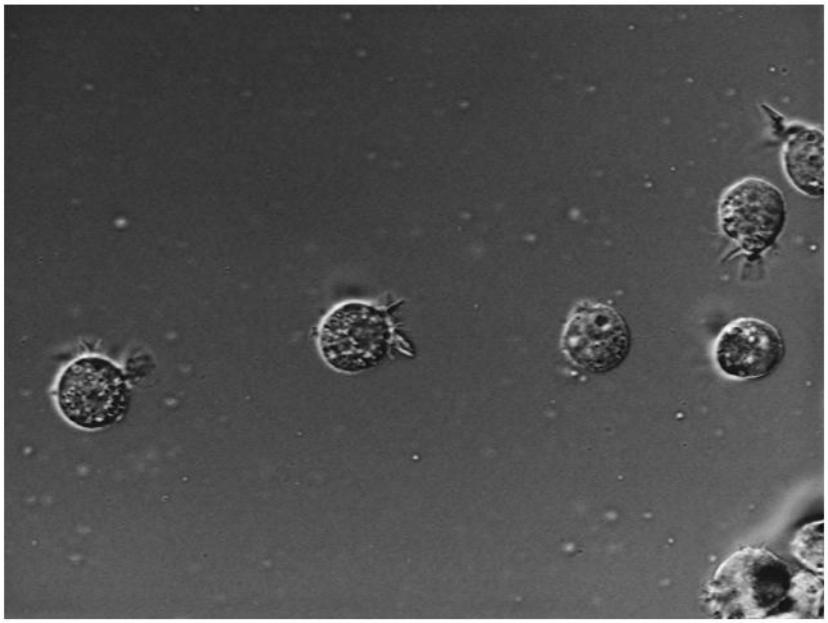
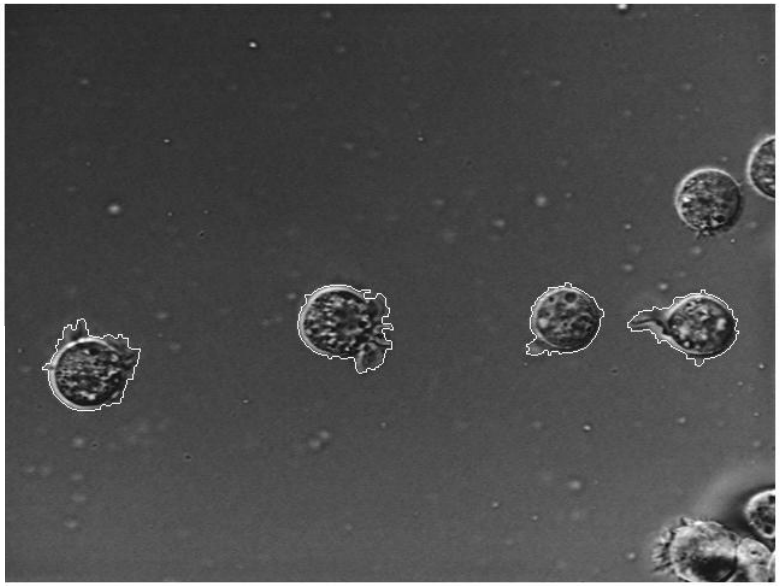
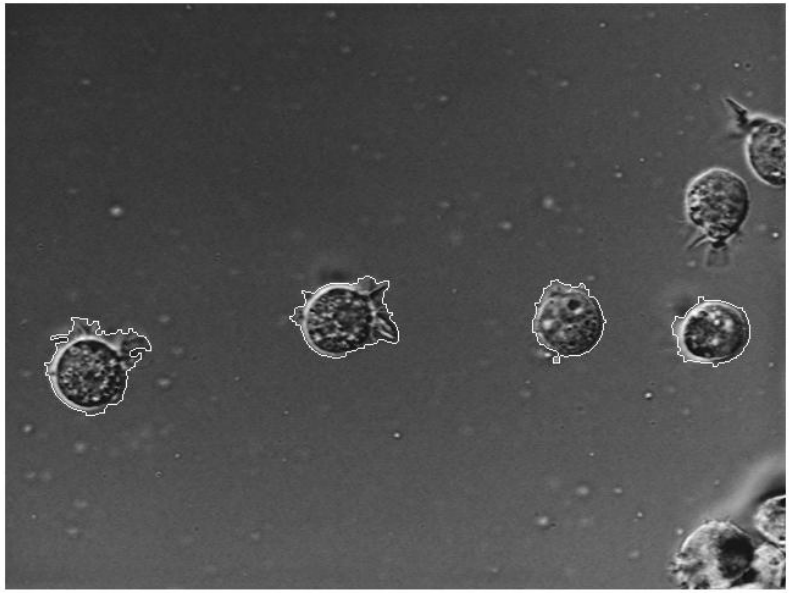
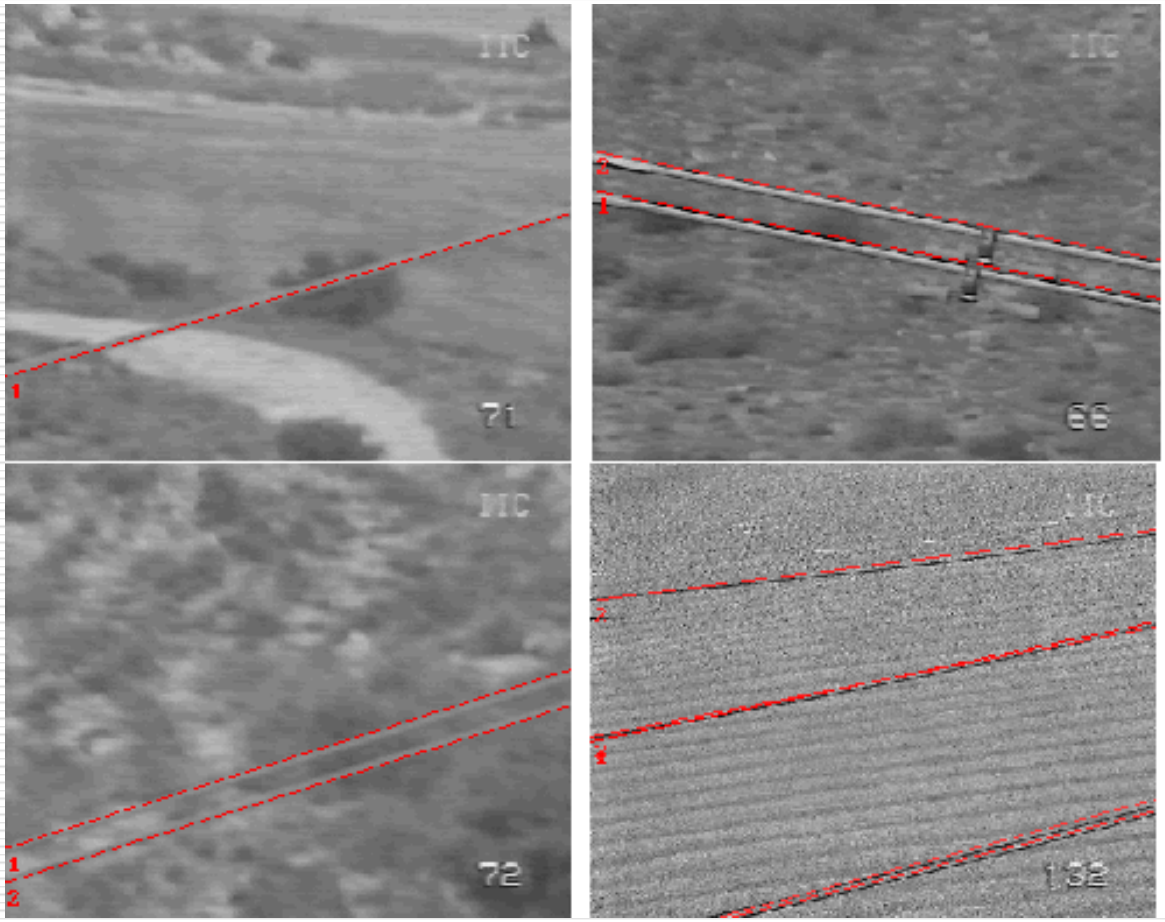


Image files courtesy of Alan Partin
Johns Hopkins University

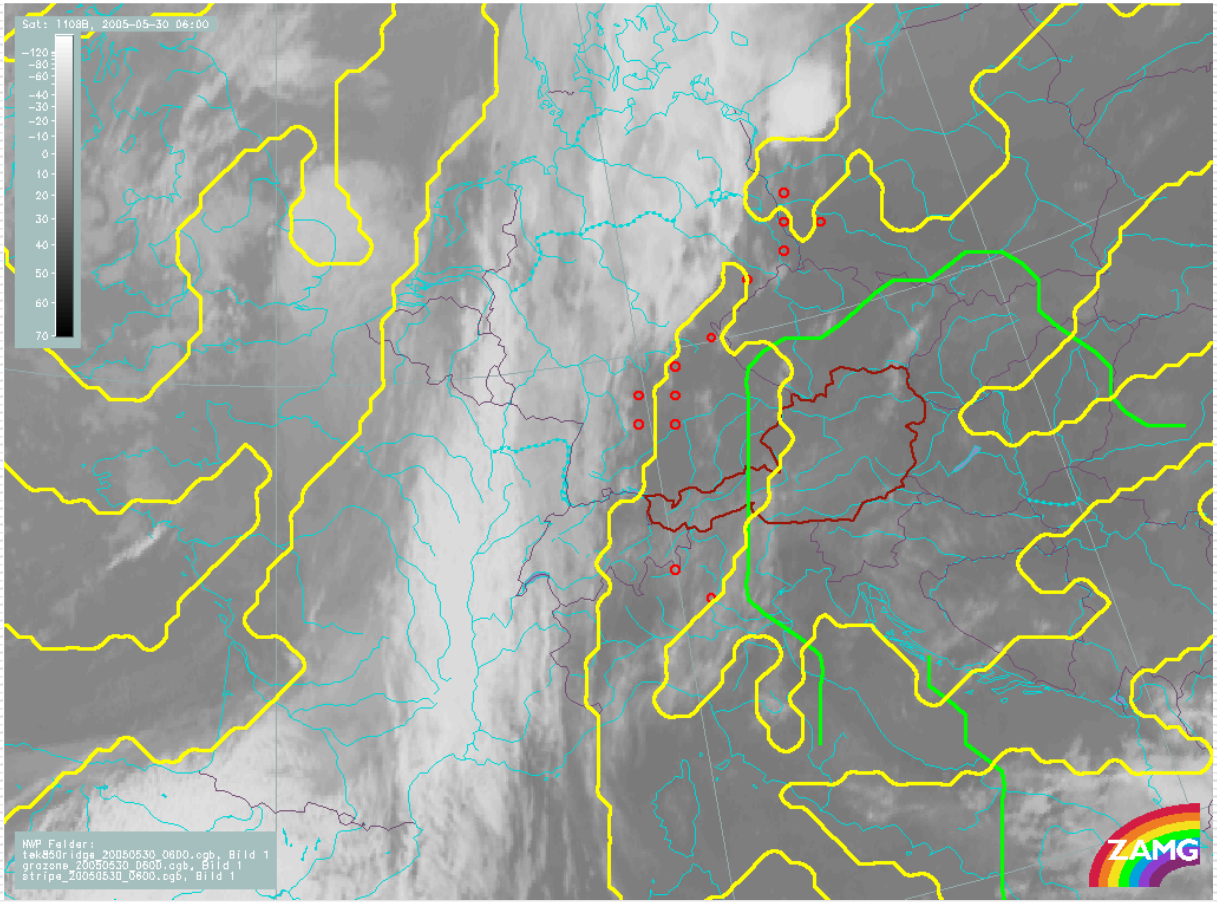




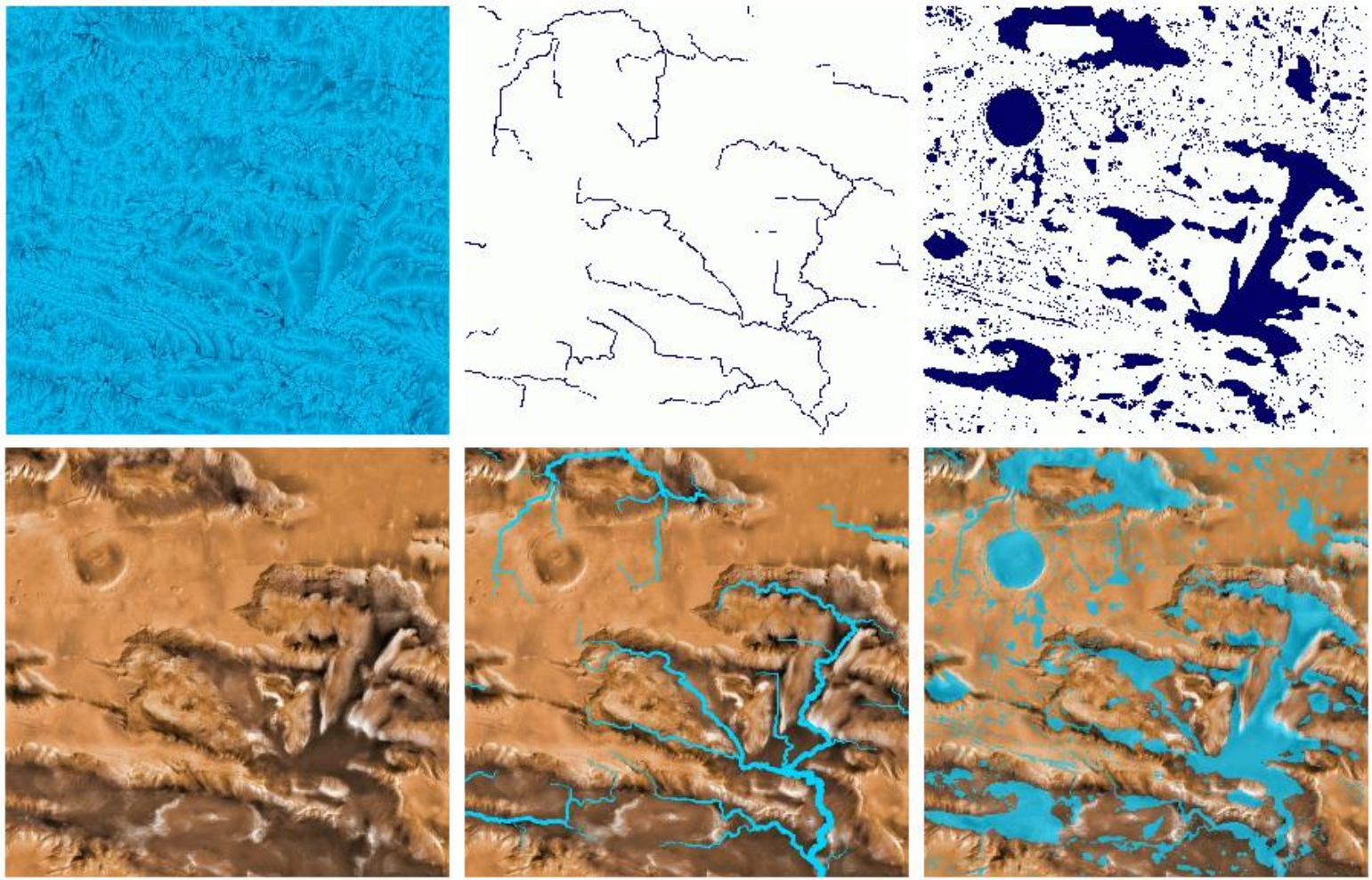
Line Detection



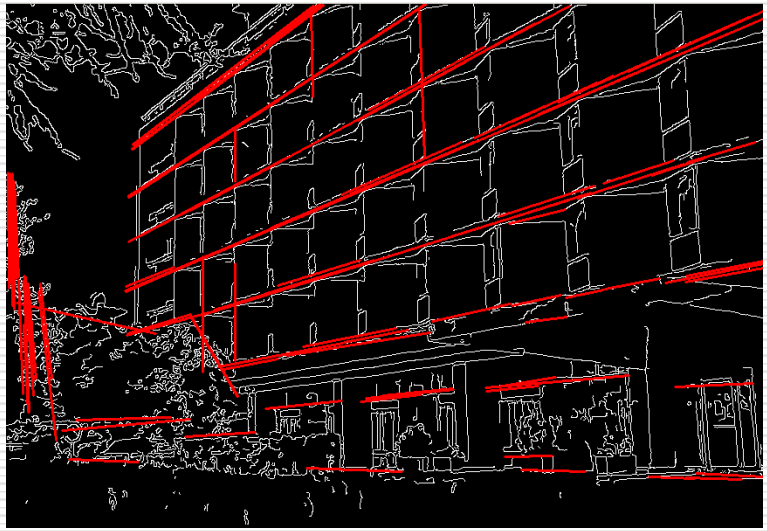
Line Detection (Con't)



Line Detection (Con't)



Line Detection (Con't)



Hough Transform

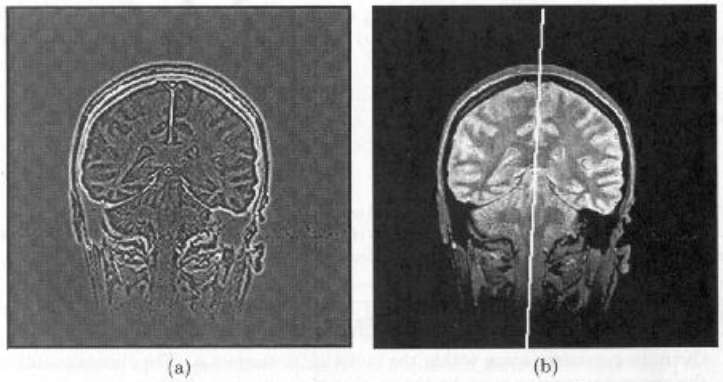


Figure 5.37: *Hough transform line detection used for MRI brain segmentation to the left and right hemispheres: (a) edge image; (b) segmentation line in original image data.*

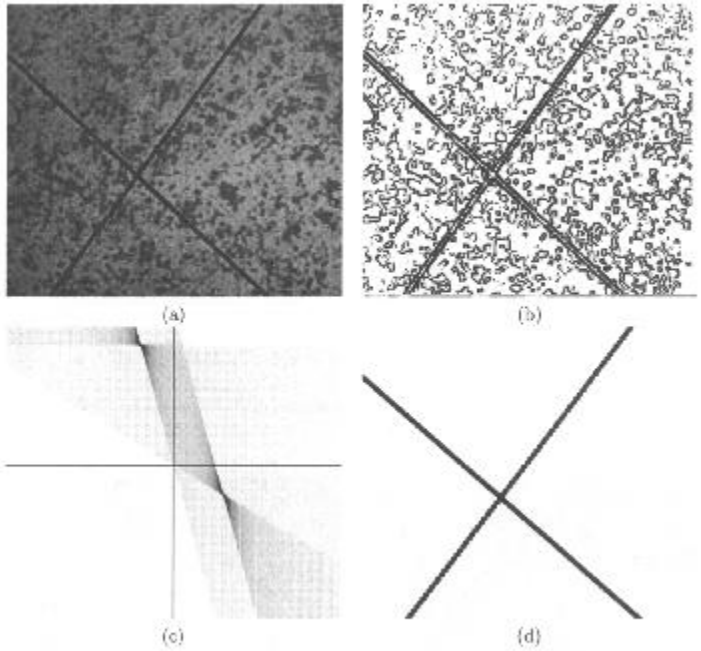
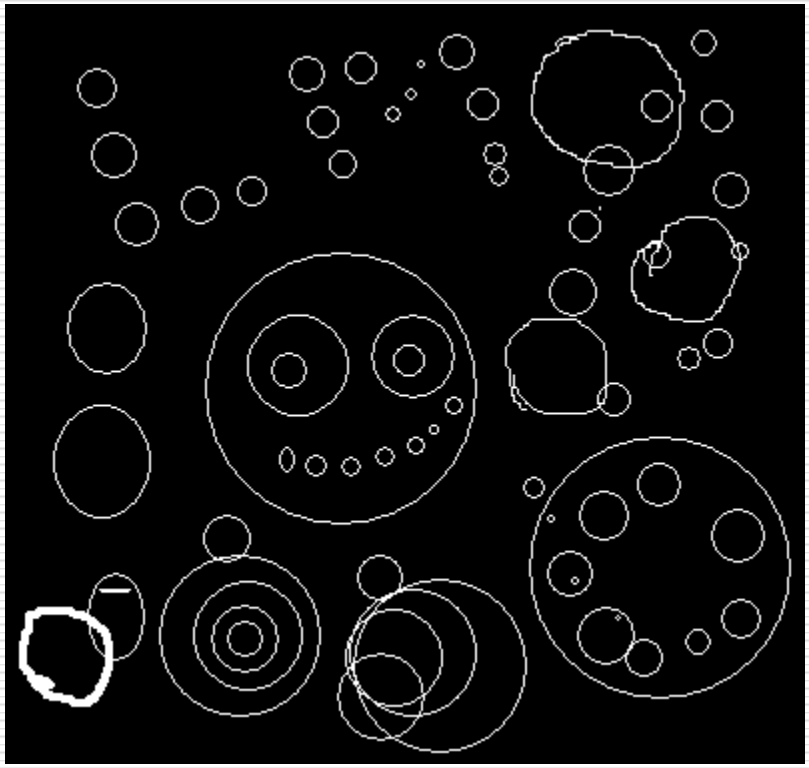
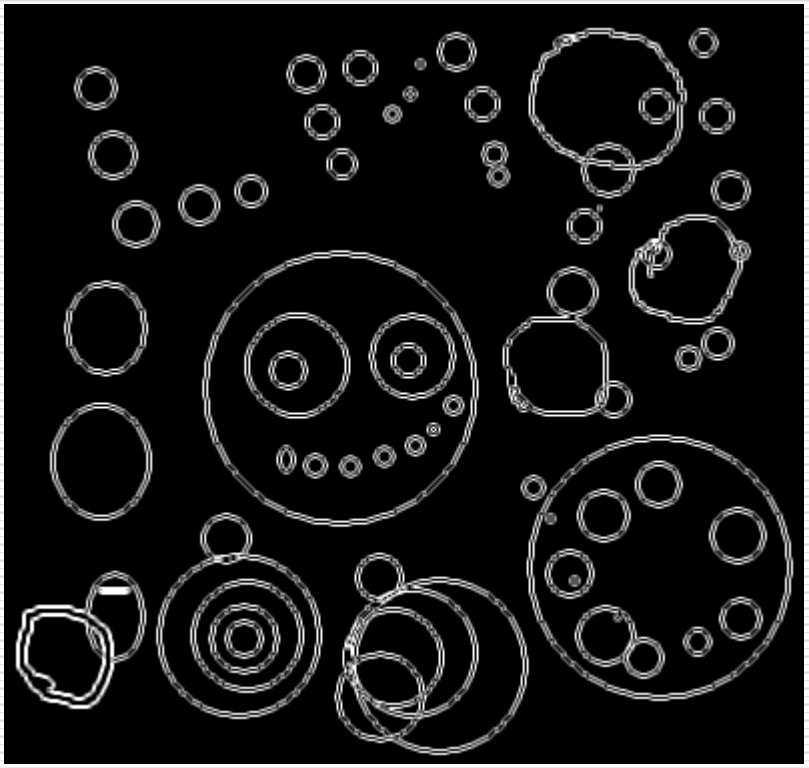


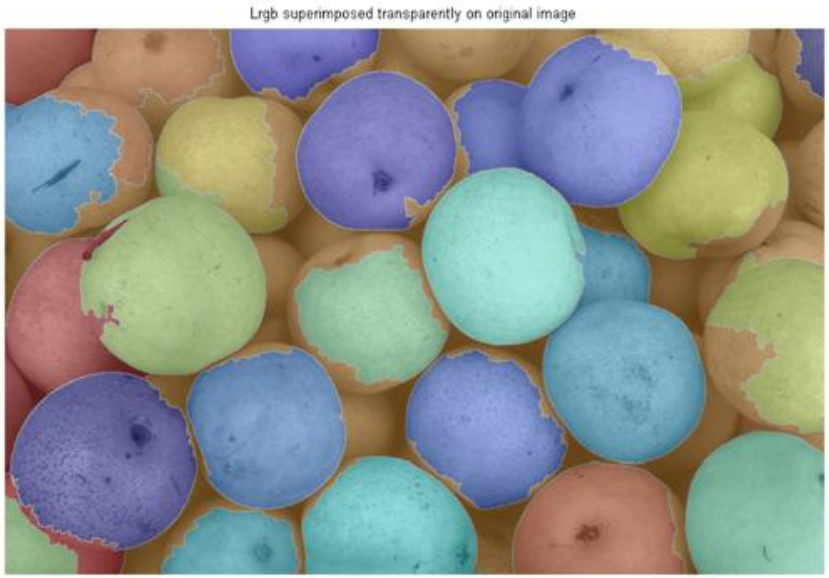
Figure 5.35: *Hough transform – line detection: (a) original image; (b) edge image (note many edges, which do not belong to the line); (c) parameter space; (d) detected lines.*

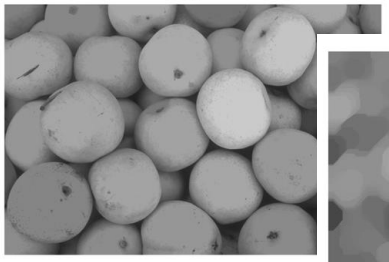
Hough Transform (Con't)

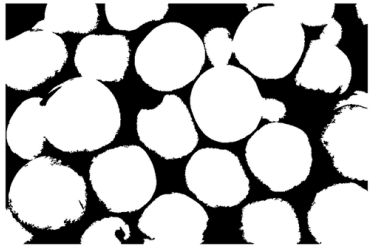


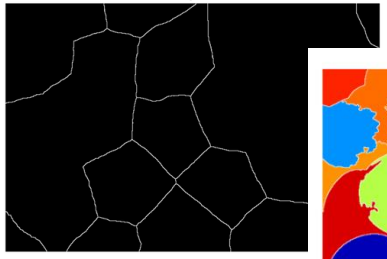


Hough Transform (Con't)









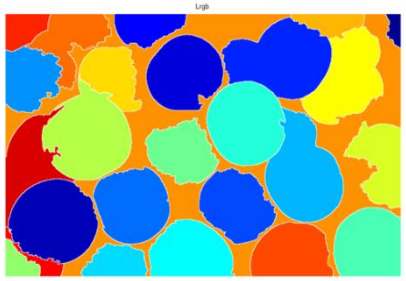


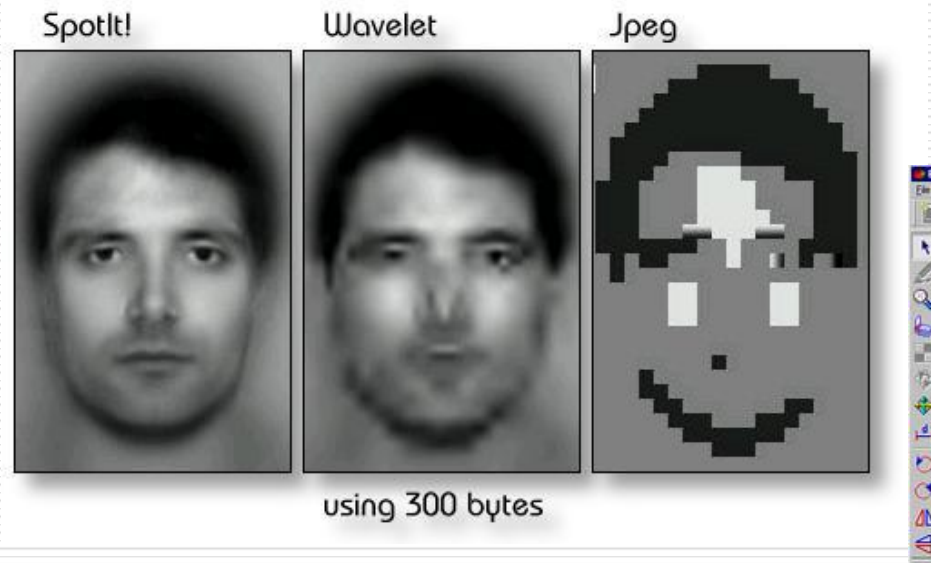
Image Compression

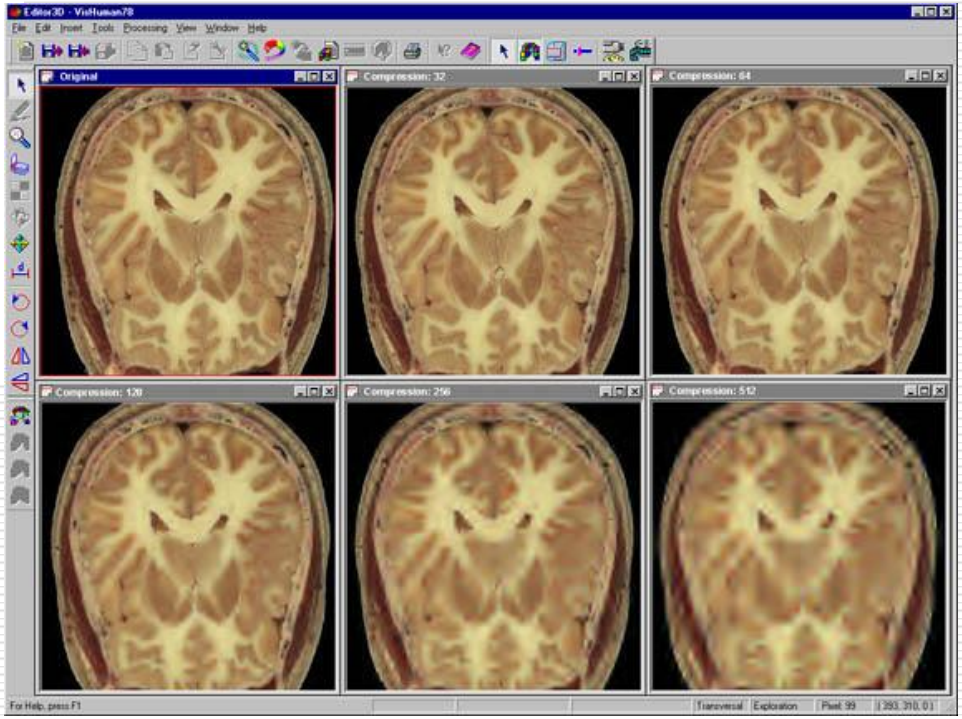
- เป็นกระบวนการที่ได้รับความสำคัญเพิ่มมากขึ้น อันเนื่องจากเป็นส่วนหนึ่งที่ช่วยในการส่งผ่านข้อมูลผ่านระบบเครือข่าย มีหลายเทคนิคที่ถูกนำมาใช้ในการบีบอัดข้อมูลภาพ เช่น Wavelet
- การบีบอัดสัญญาณ เป็นเทคนิคที่ใช้ในการลดพื้นที่ในการจัดเก็บข้อมูลภาพ หรือลด Bandwidth ที่ใช้ในการส่งผ่านเครือข่าย
- การบีบอัดข้อมูลภาพ มี 2 ชนิดคือ
  - การบีบอัดแบบไม่สูญเสียข้อมูล (Lossless compression)
  - การบีบอัดแบบสูญเสียข้อมูล (Lossy compression)

Image Compression (Con't)

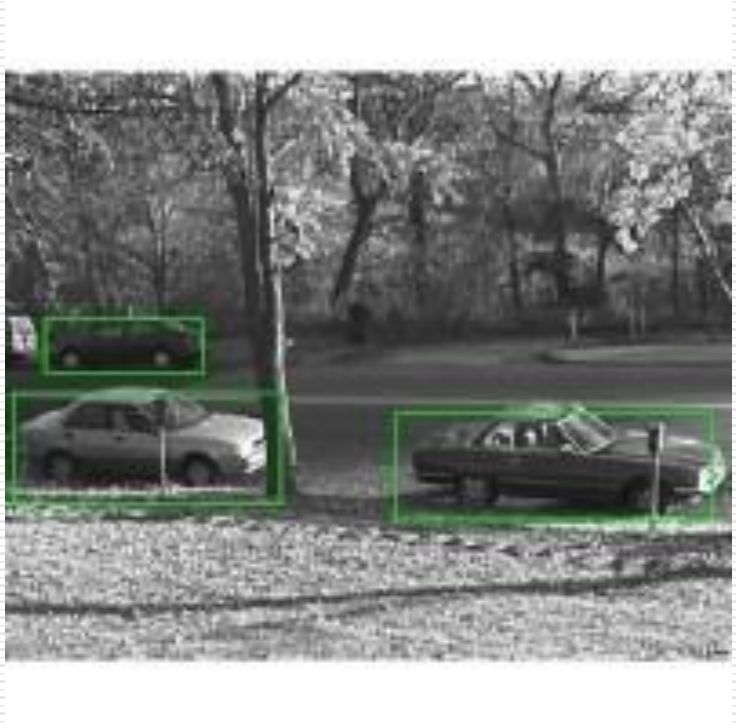
- เทคนิคที่ใช้ในการบีบอัดสัญญาณ มี 4 กลุ่มด้วยกัน
  - Statistical compression technique (Huffman coding)
  - Spatial compression technique (run length technique)
  - Quantizing compression technique (Quantize levels reducing)
  - Fractal compression technique (Most compress ratio)

Image Compression (Con't)





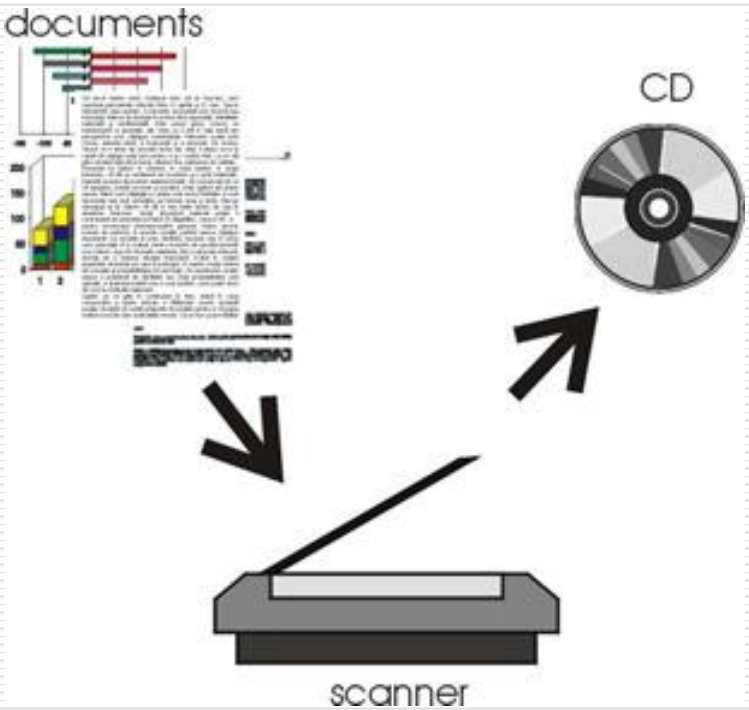
Object Recognition

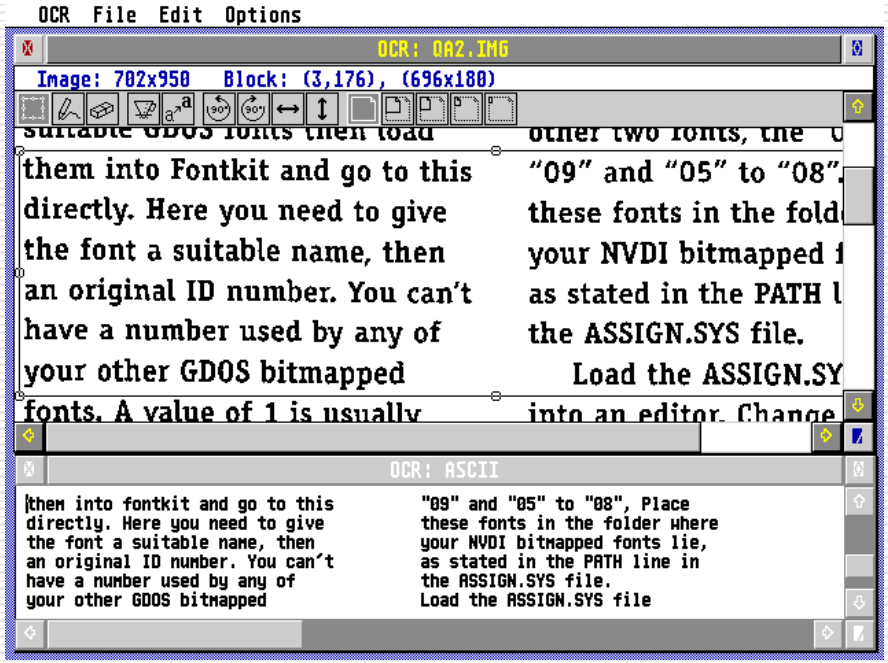


Object Recognition (Con't)

Optical Character Recognition

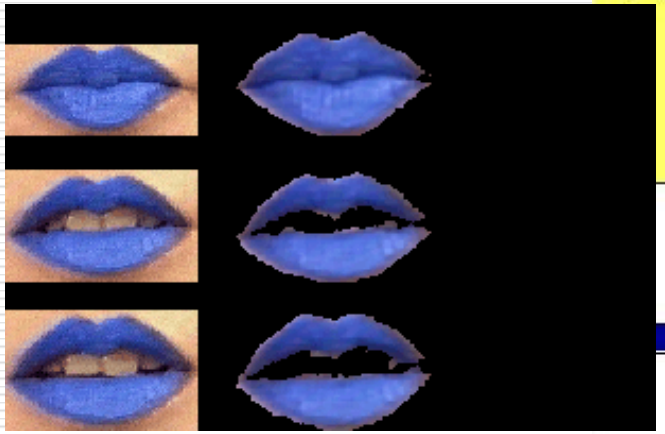
- OCR, Scanned Text Image → Text or Word.

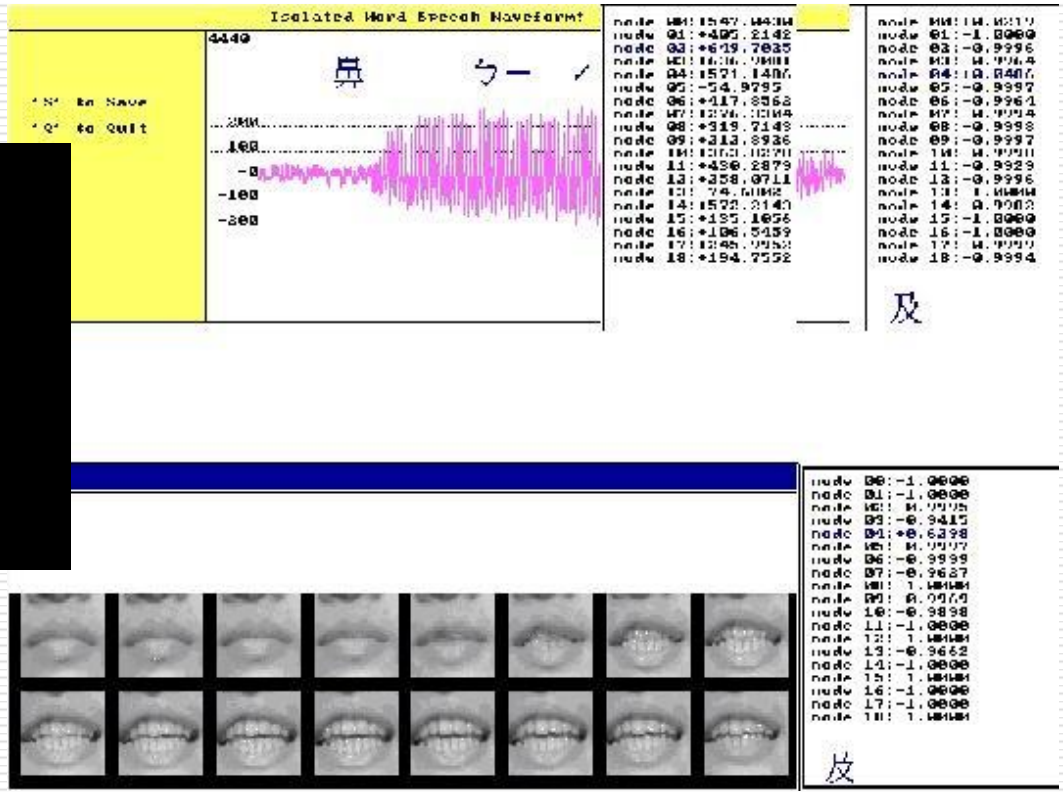




Lisp Reading Recognition

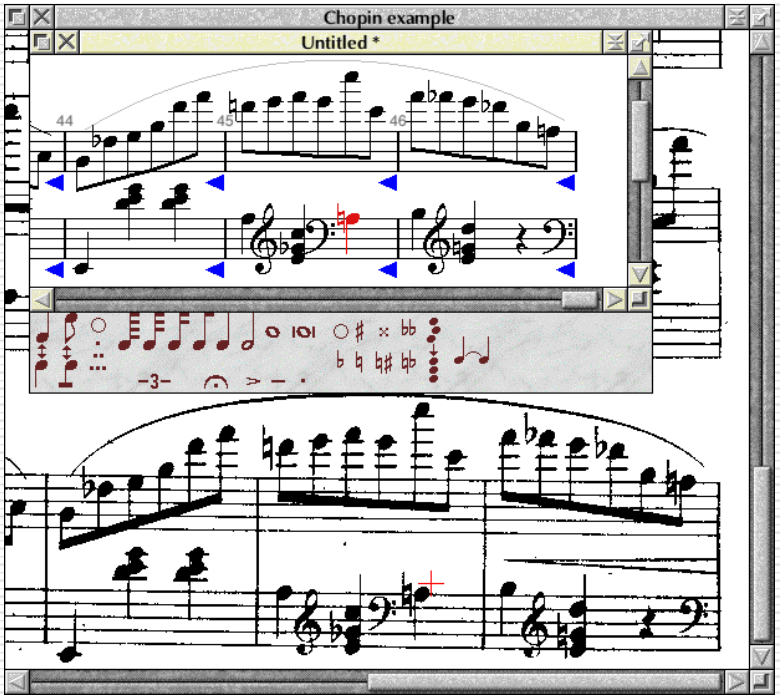
- Lisp Image → Text or Speech.

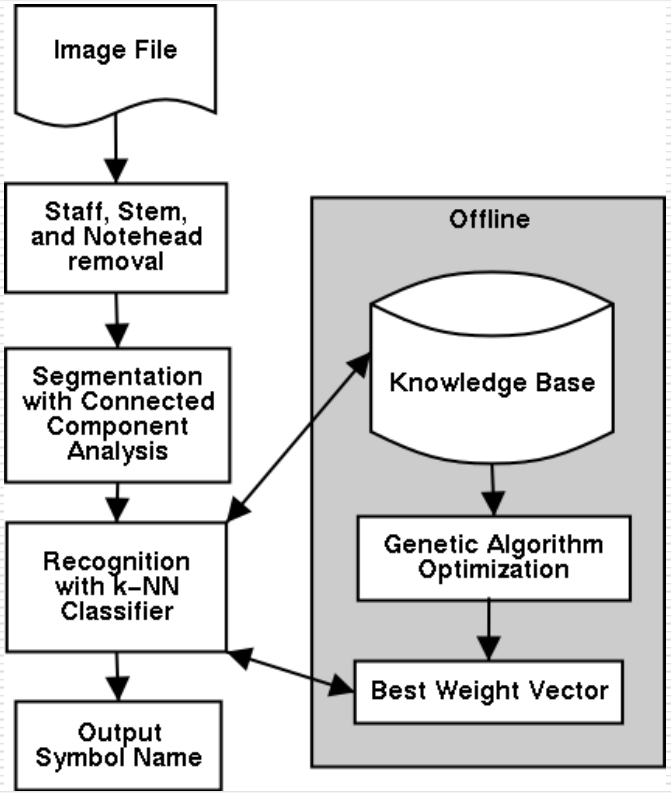




Musical Chord Recognition

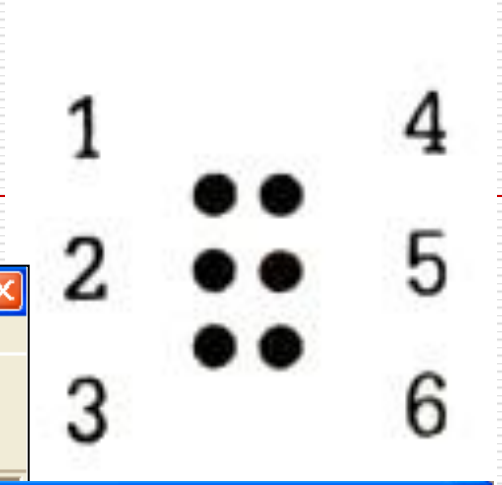
- MCR, Note Image → Note Sequence or Song or MIDI File.

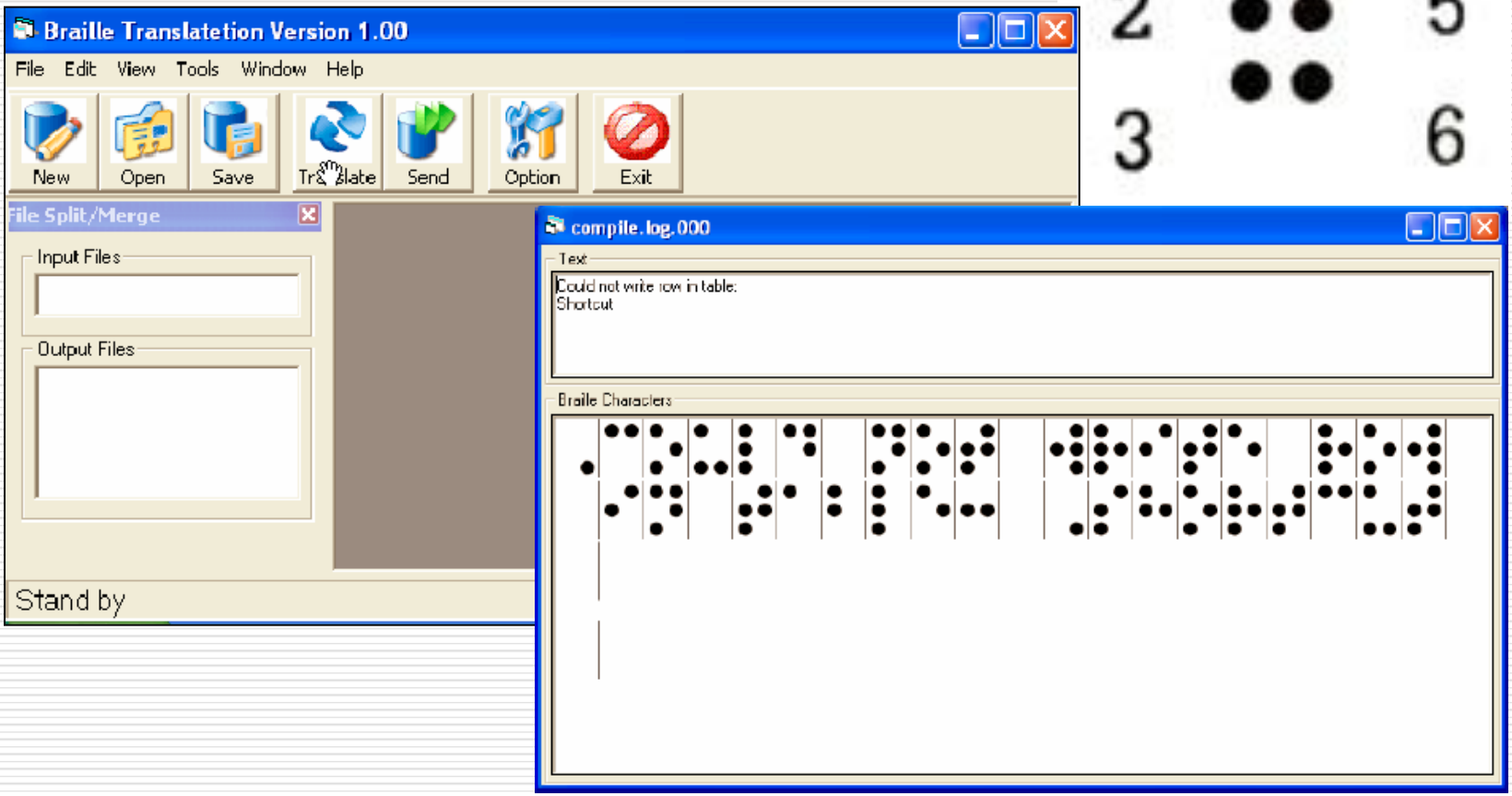




2D Barcode Recognition

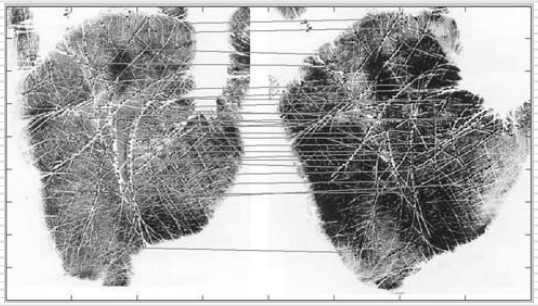
Braille Recognition



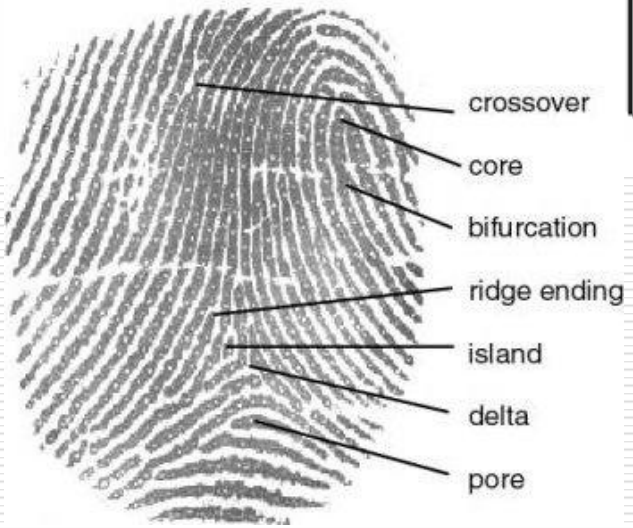


Physical Biometrics

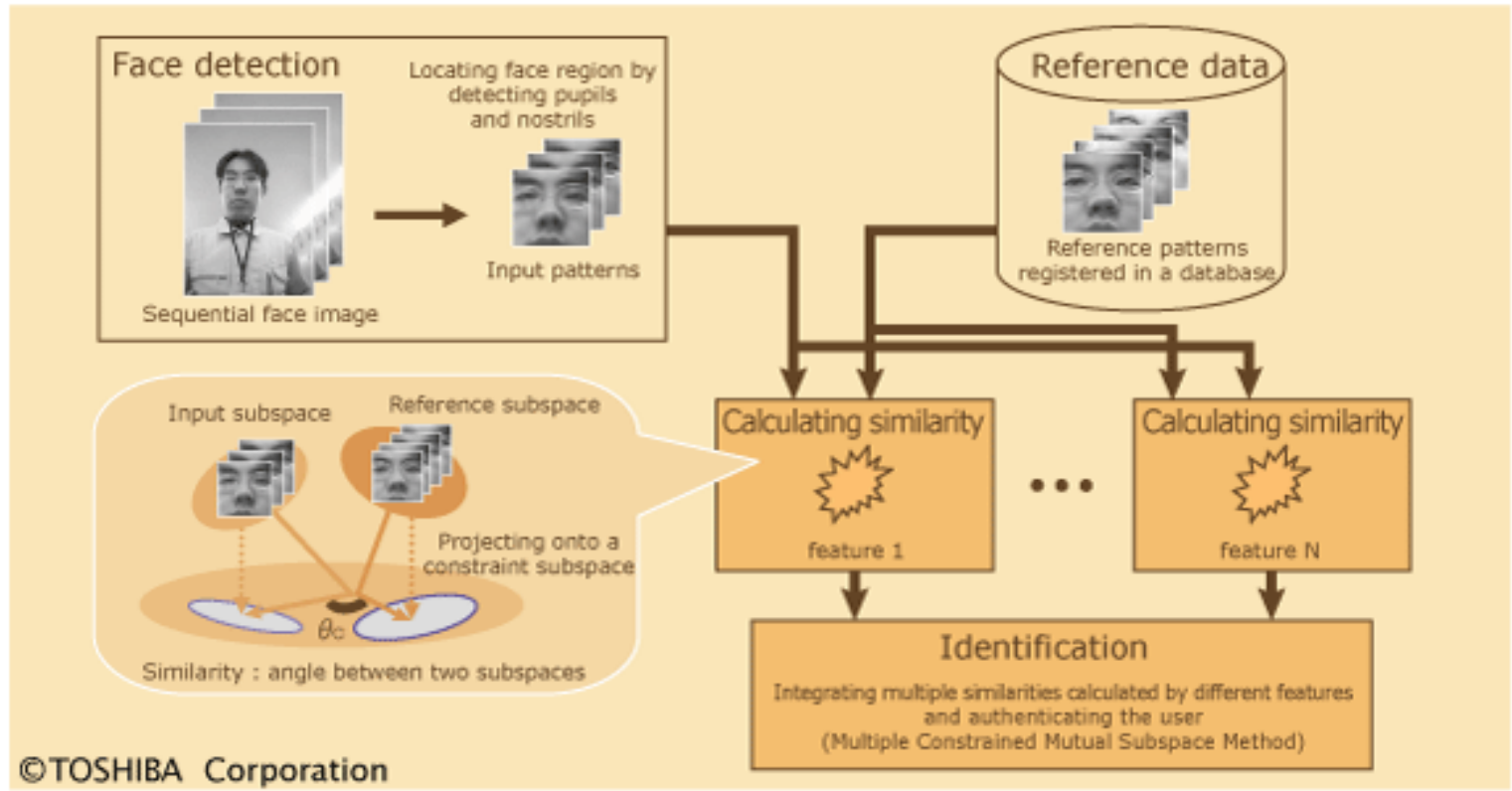
- Example
  - Iris Detection, Fingerprint
  - Hand Geometry, Face Detection
  - Palm print Detection, Body Geometry
  - Measurement Biometric
  - Dependent on environment
  - Dependent on interaction



Finger Print Recognition

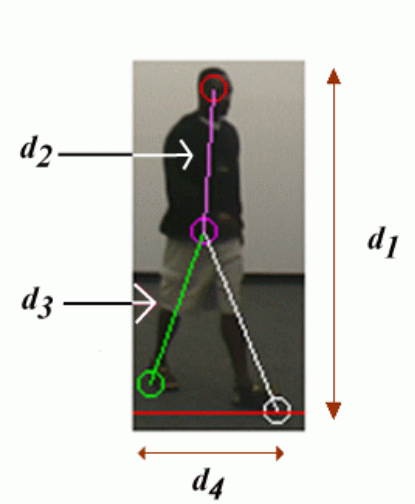


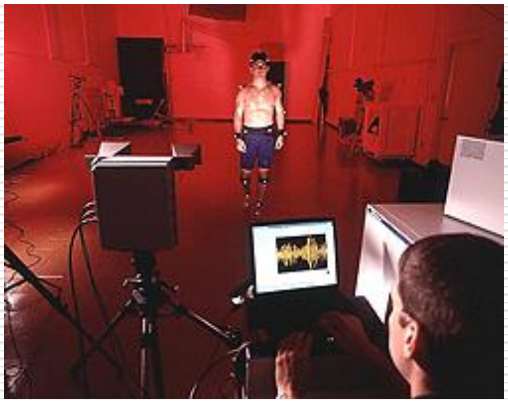
Face Recognition (Con't)



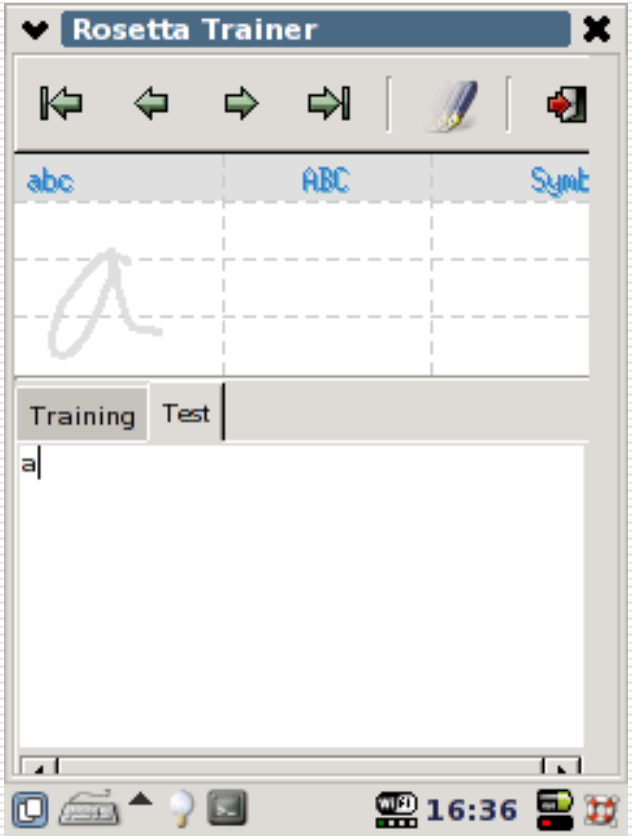
Behavior Biometrics

- **Example**
  - Handwriting, Signature, Gait
  - Performance/Temporal biometric
  - Dependent on state of mind





Handwriting Recognition



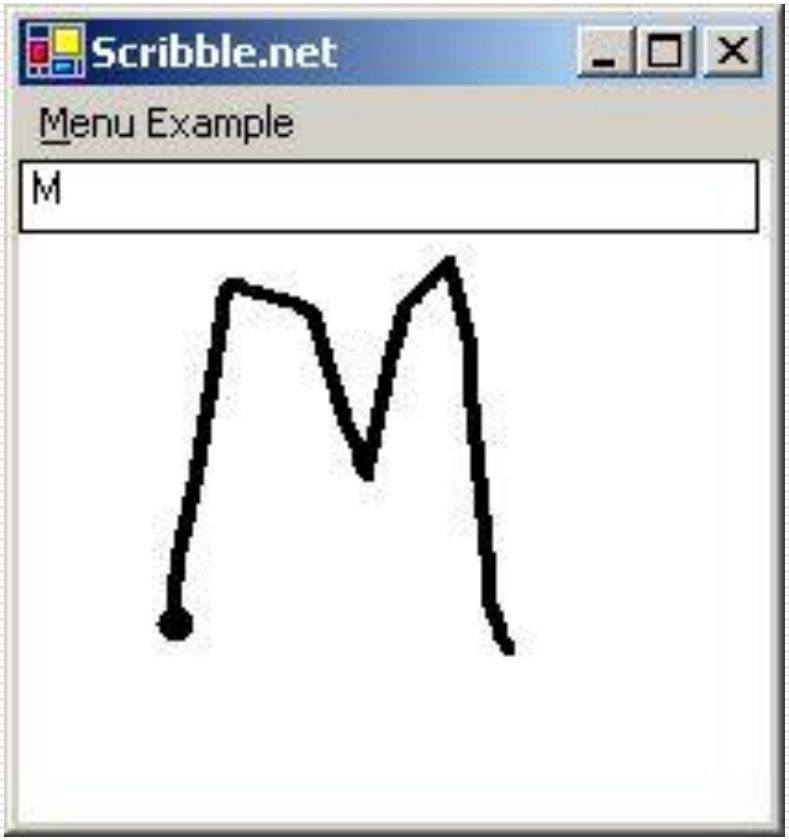


Image Retrieval System (Con't)

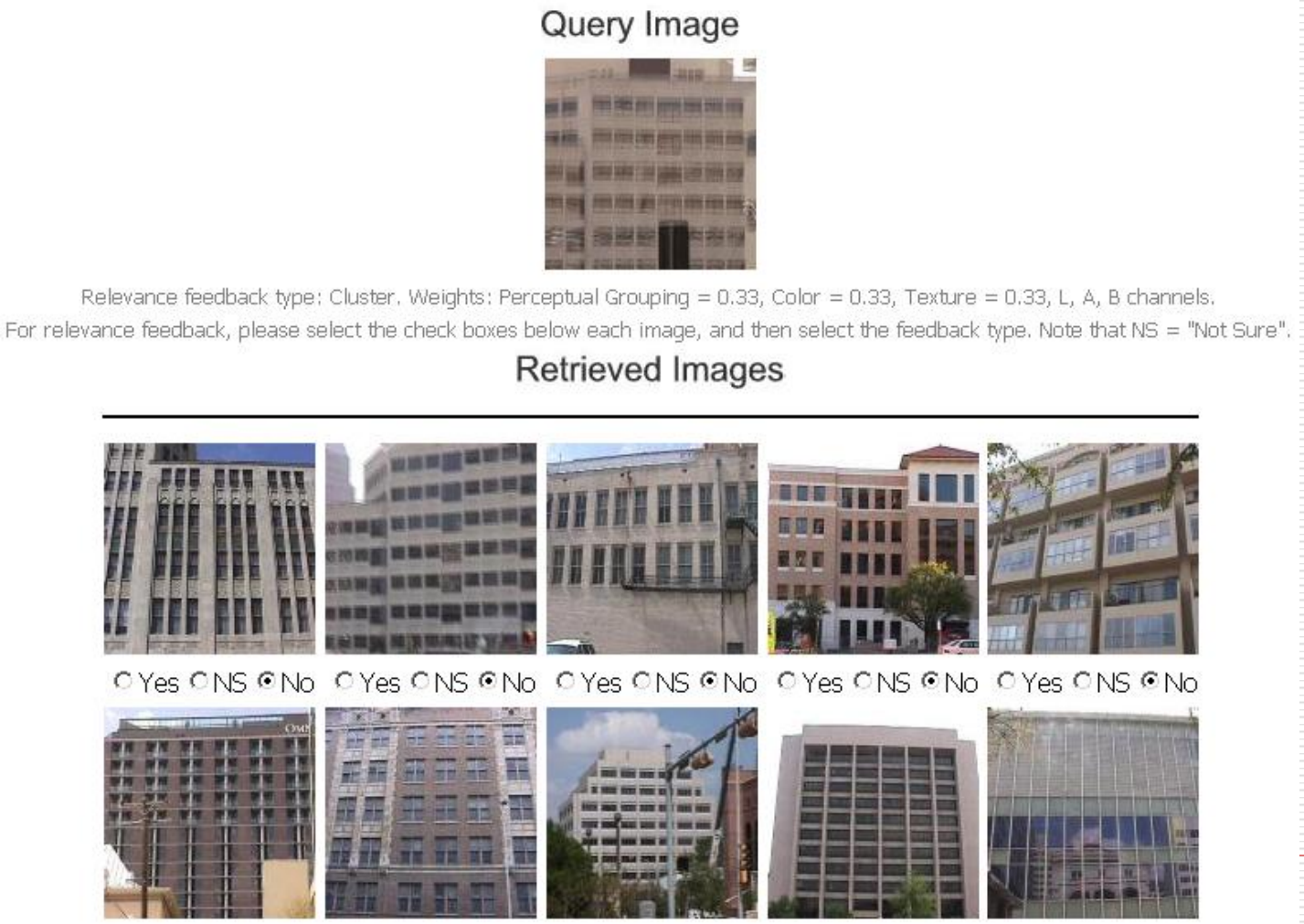
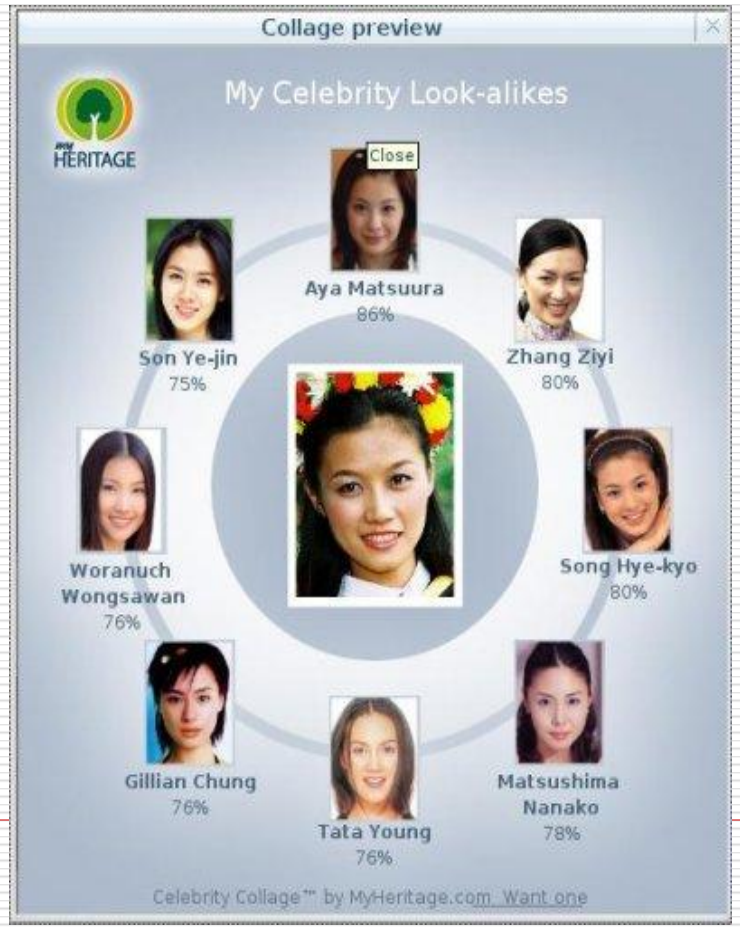
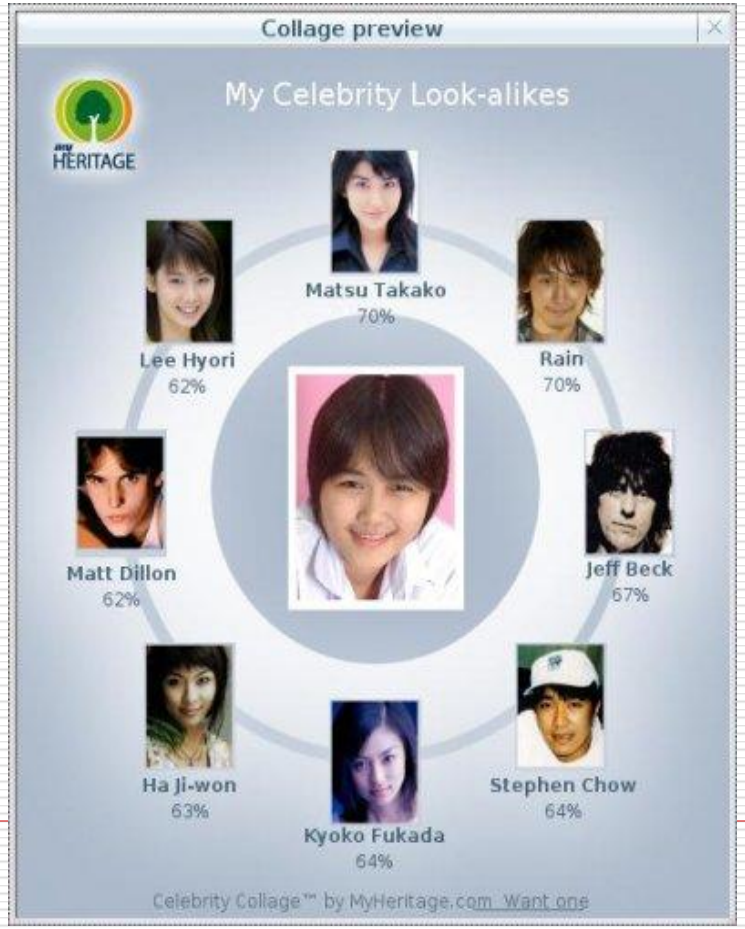


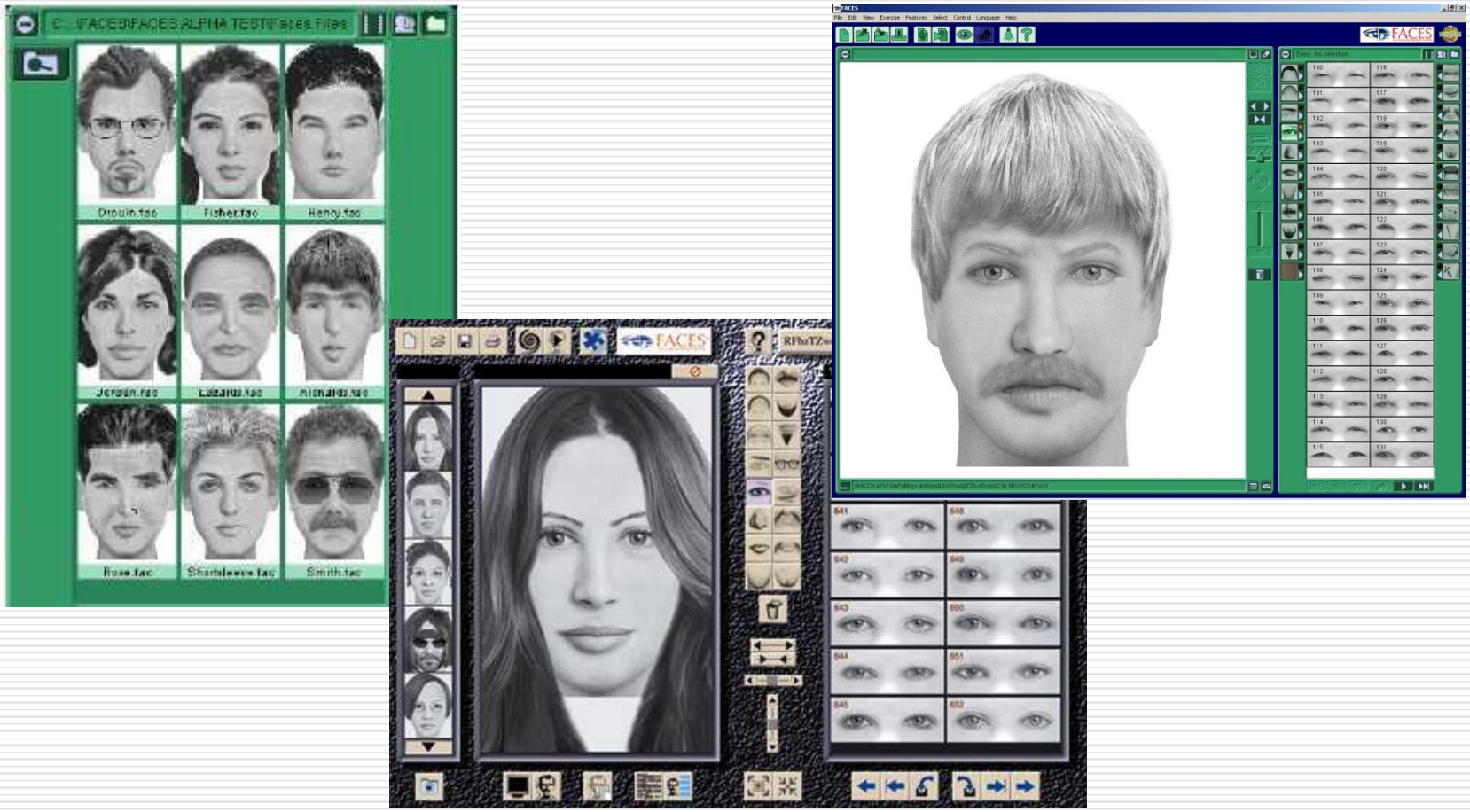
Image Similarity

- http://www.myheritage.com/





Crime-Face Retrieval



Conclusions

- DIP is methodology for Original Image transformation to another form.
- After processing → Output are image, feature vectors or sometime image sized is reduced.
- DIP is tool for Image enhancement or Image retouching.
- DIP is pre-processing for Object Recognition and Pattern Recognition.